



IFRS

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IFRS S2 - Climate-related Disclosures

ISSB

Provides guidelines for disclosing climate-related risks/opportunities.

Aligns with TCFD framework. (Task Force ON Climate Related Financial Disclosures)

Covers governance, strategy, risk management, and metrics.

Report on Climate-related Disclosures

Get Ready

- Current position

- Sustainability roadmap

- Good quality data is available.

- Finalising with standards

- Educate your organisation.

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IFRS S1 - General Requirements

June 2023. → Impact / Imposed 1 Jan 24

Focuses on sustainability-related financial disclosures.

Emphasizes transparency, relevance, and consistency.

Covers governance, strategy, risk management, and metrics/targets.

Issued by ISSB

International Sustainability

Standard Board

Opportunity framework

Cash Flows

Finance

Cost of Capital

Short

Medium

Long term

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IFRS 5 - Non-current Assets Held for Sale

1 Jan 2005

Deals with classification, measurement and presentation of non-current assets held for sale and discontinued operations.

Assets must be available for immediate sale and sale must be highly probable.

Depreciation

Carrying Amount

Impairment

Loss/Gain

Disposal of Asset

Statement of Comprehensive Income

Statement of Financial Position

Balance Sheet

Disclosure group

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IFRS 7 - Financial Instruments: Disclosures



Requires disclosure of significance of financial instruments.
Details on risk exposure: credit, market, and liquidity risk.
Encourages transparency for financial instrument impact.

↳ Qualitative Disclosure →
↳ Mgmt objectives Policies External
↳ Quantitative " → Managing Risk

↳ Summary data.

↳ Credit.
↳ Liquidity.

↳ Market Risk ✓

IFRS 8 - Operating Segments



Objective:

To disclose information to help users evaluate the nature and financial effects of business activities and the economic environments in which the entity operates.

Key Requirements:

- Segment disclosures based on internal management reports ✓
- Identify reportable segments using 'management approach'
- Disclose factors used to identify segments (e.g., geography, products)

Purpose for Investors:

- Understand how the business is structured and managed
- Assess performance and risks across segments

↳ Stakeholders.
↳ Decision maker.

↳ Revenue
↳ Profit.

Product & Services.

IFRS 13 - Fair Value Measurement



↳ Non financial Assets.

Key Principles:

Fair Value (FV) Definition: Exit price in an orderly transaction (market-based, not entity-specific).

Fair Value Hierarchy:

- Level 1: Quoted prices (active markets) ✓
- Level 2: Observable inputs (similar assets) ✓
- Level 3: Unobservable inputs (entity assumptions) ✓

Example:

A pharmaceutical company values an in-process R&D project (no market data). It estimates:
Future drug sales (unobservable revenue projections).
Probability of regulatory approval (management's assumption).

Discount rate (adjusted for R&D risk).

Valuation Techniques:

Market, Income (DCF), or Cost Approach.

Scope:

Applies when IFRS requires/permit FV measurement.
Covers assets, liabilities, and equity instruments.

Financial Mgmt

Level 1 – Quoted Prices (Most Reliable)



- Inputs: Quoted prices in active markets for identical assets/liabilities.
- Example: Listed equity securities on a stock exchange.
- Reliability: Highest — directly observable market prices.

Level 2 – Observable Inputs (Other than Quoted Prices)



- **Inputs:** Observable market data for similar assets/liabilities, or quoted prices in inactive markets.
- **Example:** Corporate bonds valued using market yields of similar bonds.
- **Reliability:** Moderate — based on market data but requires adjustments.

Level 3 – Unobservable Inputs (Least Reliable)



- **Inputs:** Unobservable, based on management assumptions.
- **Example:** Private equity investment valued using discounted cash flow model.
- **Reliability:** Lowest — high subjectivity.
- **Requirement:** Disclose sensitivity analysis — how valuation changes if key assumptions (e.g., discount rate, growth rate) vary.

IFRS 15 - Revenue from Contracts



Outlines principles for recognizing revenue from customer contracts.

5-step model: Identify contract, performance obligations, transaction price, allocation, recognition.

Ensures consistency across industries.

The 5-Step Model for Revenue Recognition



Step 1: Identify the Contract

Criteria:

- Approved by parties, enforceable rights/obligations.
- Payment terms identifiable.
- Commercial substance (i.e., cash flow impact).
- Probable collection of consideration.

Step 2: Identify Performance Obligations

Distinct goods/services promised to the customer (e.g., phone + warranty = 2 obligations).

"Distinct" means:

- Capable of being separate (customer can benefit alone or with other resources).
- Separately identifiable in the contract.

The 5-Step Model for Revenue Recognition



Step 3: Determine Transaction Price

Includes:

Fixed/variable consideration (e.g., discounts, rebates, bonuses).

Adjusted for time value of money if payment is deferred (>1 year).

Variable consideration (e.g., royalties) estimated using the most likely amount or expected value.

Step 4: Allocate Price to Performance Obligations

Based on standalone selling prices (if not observable, use estimation methods like cost-plus or market comparables).

Step 5: Recognize Revenue When Obligations Are Satisfied Over time (if one of these applies):

Customer controls asset as it's created (e.g., construction project).

Entity's work creates/enhances an asset the customer controls.

No alternative use + right to payment for progress.

Key Practical Applications



Software/SaaS: License fees vs. service revenue (e.g., upfront license + updates).

Construction: Percentage-of-completion method (over time).

Retail: Revenue recognized at delivery (point in time).

Disclosure Requirements

Quantitative: Revenue streams, contract balances (e.g., deferred revenue).

Qualitative: Judgments (e.g., variable consideration estimates).

Common Challenges

Variable consideration: Estimating refunds, bonuses, or penalties.

Distinct obligations: Bundled goods/services (e.g., phone + cloud storage).

Timing: Over time vs. point-in-time decisions.

(10% Advance → 20%

20% → 30%

50% → 5%

→ 35% (Full final payment is done)

IAS 16 - Property, Plant and Equipment



Prescribes accounting treatment for PP&E.

Includes recognition, measurement at cost, depreciation, and derecognition.

Allows cost model or revaluation model post recognition.

Valuation expert

M&A

Key Practical Applications



Practical Implications & Challenges

Critical Considerations:

Non-Financial Assets: Measured at "highest and best use."

Liabilities/Equity: Includes non-performance risk (e.g., credit risk).

Disclosures: Must disclose Fair Value hierarchy level and Level 3 sensitivity.

IAS 2 - Inventories



Provides guidance on accounting for inventories.

Inventories measured at lower of cost and net realizable value.

Cost includes purchase, conversion, and other costs.

IAS 2 versus US GAAP



Topic	IAS 2	U.S. GAAP
LIFO	☒ Banned	☒ Allowed
Write-downs	☒ Reversible	☒ Irreversible

107 ✓ 500000 Asset Value
50000 Dep
450000 Book Value
450000 NRV
405000 50000

Objectives of IAS 2



Purpose: Standardize inventory accounting under IFRS.

Scope: *work in progress*

Finished goods, WIP, raw materials ✓

Excludes: Financial instruments, biological assets. ✗

Initial Measurement: ✗

At cost (purchase price + conversion costs + other costs). ✓

Subsequent Measurement: ✗

Lower of cost or Net Realizable Value (NRV). ✓

Formula: ✗

NRV = Selling price - Completion/selling costs. ✓

Allowed Methods: ✓

FIFO (First-in, First-out). ✓

Weighted Average Cost. ✓

Specific Identification. ✓

Prohibited: LIFO. ✓

US GAAP vs IFRS - R&D



US GAAP: R&D expensed as incurred (except software).

IFRS: Research expensed; Development capitalized if criteria met.

IFRS encourages capitalization leading to higher assets/profits.



12 Red Flags – Accounting and Finance

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Red Flag 3 - Accounts Receivable Growth > Sales Growth



Sign: Accounts receivable rising faster than sales growth.

Risk: Could mean relaxed credit terms leading to potential defaults, or 'channel stuffing' to inflate current sales.

Example: Pushing excess goods to distributors before quarter-end to record revenue.

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Red Flag 2 - Transactions Boosting Profits Without Real Performance



Sign: Unusual asset sales, debt-for-equity swaps, or other one-off transactions that inflate reported profits.
Risk: Creates a misleading picture of operational success.

Example: Selling a building to realize a gain when core business is underperforming.

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Red Flag 4 - Inventory Build-up > Sales Growth



Sign: Inventory increasing disproportionately compared to sales.

Risk: Finished goods build-up could mean slowing demand → possible price cuts or write-downs.

Note: Increase in WIP may indicate expected demand; raw material build-up may signal procurement inefficiencies.

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Red Flag 5 - Earnings-Cash Flow Gap Increasing



Sign: Reported net income rising, but cash flows from operations stagnant or falling.

Risk: Aggressive accrual estimates or revenue recognition methods.

Example: Using % of completion method more aggressively without actual cash inflow.

Red Flag 6 - Earnings -Taxable Income Gap Increasing



Sign: Growing difference between book income and taxable income without changes in tax laws.

Risk: May suggest aggressive accounting choices for financial reporting.

Example: Underestimating warranty expenses for financial reports while reporting them accurately for tax purposes.

Red Flag 7 - Complex Financing Structures



Sign: Use of R&D partnerships, Special Purpose Entities (SPEs), or selling receivables with recourse.

Risk: Potential understatement of liabilities or overstatement of assets.

Example: Enron's SPE structures used to hide debt.

Red Flag 8 - Large Asset Write-offs



Sign: Sudden, unexpected impairments or write-downs.

Risk: Management may have delayed recognition of poor performance.

Example: Writing off an overvalued acquisition.

Red Flag 9 - Large 4th-Quarter Adjustments



Sign: Significant year-end adjustments following audit.
Risk: Interim financial statements were overly optimistic.
Example: Making large provisions for bad debts only at year-end.

Red Flag 10 - Auditor-Related Concerns



Sign: Qualified opinions, unexplained changes in auditors.
Risk: Management may be 'opinion shopping' for favorable auditors.
Example: Switching auditors after disputes over revenue recognition.

Red Flag 11 - Related-Party Transactions

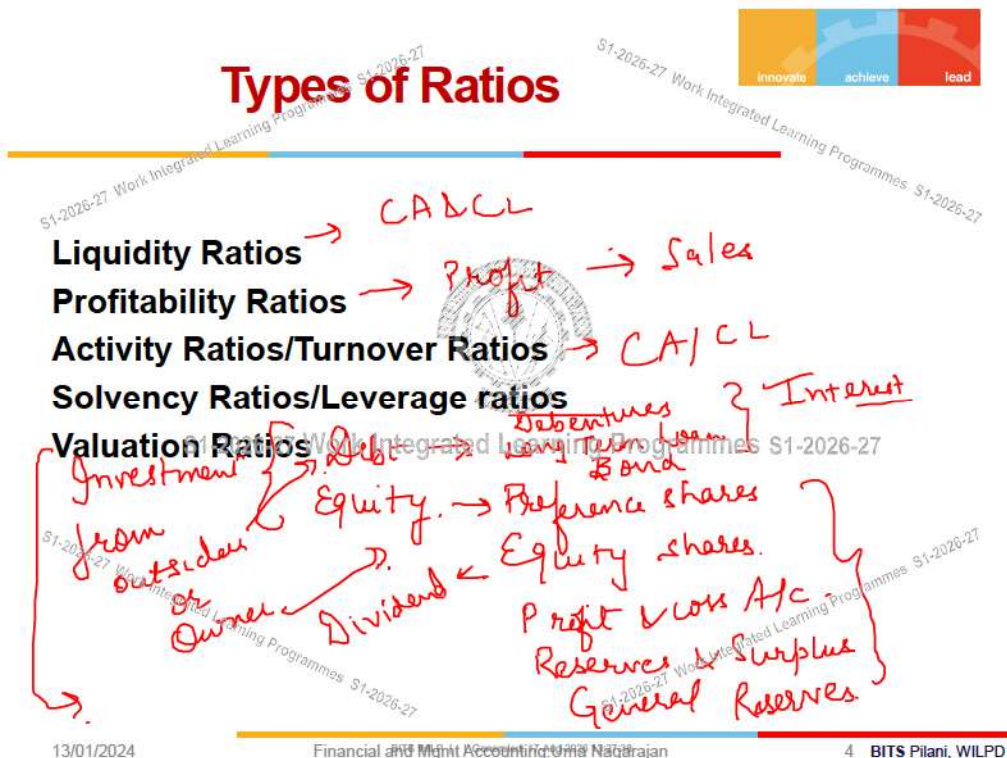
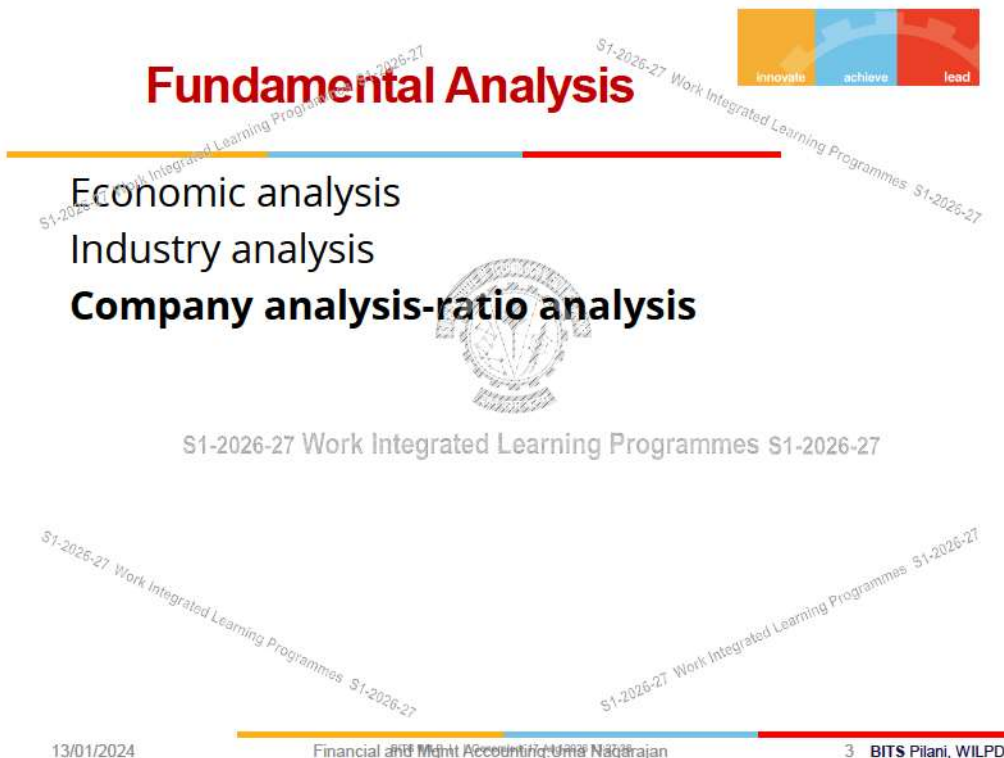
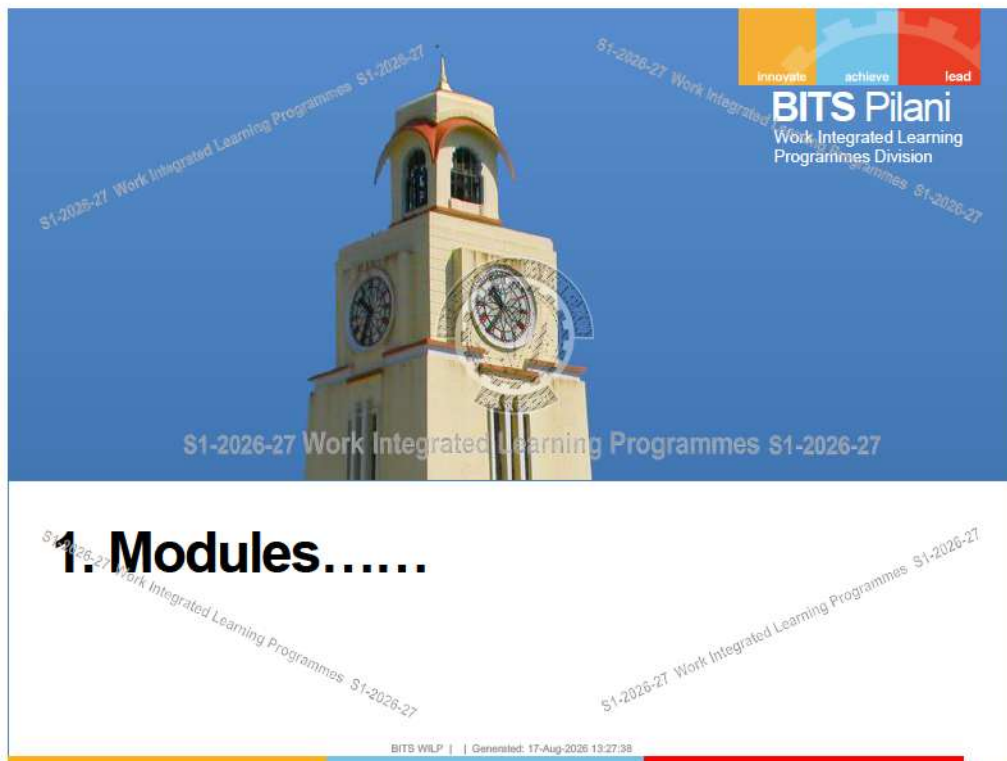
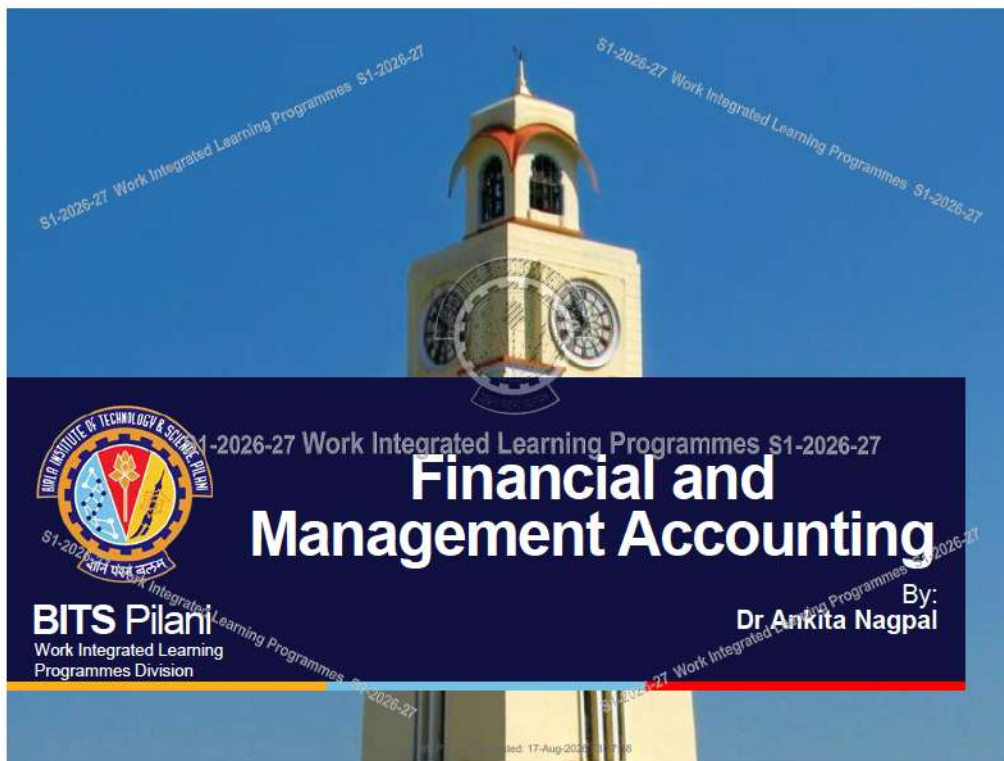


Sign: Deals between company and affiliated entities or insiders.
Risk: Non-arm's-length terms; estimates may be biased.
Example: Selling products at inflated prices to a sister company to boost revenues.

Red Flag 12 - Off-Balance Sheet Items & Contingencies



Sign: Unexplained growth in guarantees, commitments, or other off-balance sheet items.
Risk: Concealing debt or risks to make financial position look better.
Example: Large operating lease commitments not recorded as liabilities.



Liquidity Ratios



Current Ratio=

Current Assets/Current Liabilities

2: 1 ✓

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Liquidity Ratios



Liquidity Ratios



Current Ratio=

Current Assets/Current Liabilities

Current assets =

Any asset that can be converted into cash in less than a year →

Cash, Bank, stock inventory, Receivables, short term securities, Prepaid, Accrual - etc.

Current liabilities:

Any liability that needs to be paid within a year

Payables, short term loan; Bank overdraft; Outstanding Expenses

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Liquidity Ratios



Current assets =

Any asset that can be converted into cash in less than a year

Cash in hand/at bank

Fixed deposits(current investments/ST investments)

Stock/Inventories (RM+WIP+FG)

Debtors/Bills receivable

o/s incomes

Prepaid expenses



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Current liabilities

ST loan – due in a year

Creditors/bills payable

Bank over draft

o/s expense

Prereceived income



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takeaway



Liquidity ratios

Current ratio = CA/CL

Liquid ratio/quick ratio = (CA – prepaid exp – inventory)/CL

Absolute liquid ratio = CASH+BANK+ST INVESTMENTS

/CL

Quick Assets = CA – prepaid Exp – Inventory.

Liquidity Ratios



Current Ratio=

Current Assets/Current Liabilities

Liquid Ratio/Quick Ratio/Acid Test Ratio=

Liquid Assets/Current Liabilities

Liquid assets = CA – inventory – prepaid expenses.

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Liquidity Ratios



Absolute Cash Ratio/Absolute liquid ratio =

Absolute Cash/Current Liabilities

Abs cash = Cash in hand + cash at bank + st investments

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Current Liabilities

Short Tem Loans

Bank Overdrafts

Creditors

Payables

Outstanding Expenses

Incomes Received in Advance

Current Assets

Cash

Bank

Current investments/Marketable Securities

Debtors

Receivables

Inventory / Stock

Prepaid Expenses

Accrued Incomes

Absolute Liquid Assets

Quick Assets Liquid assets

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Profitability Ratios



Gross Profit Ratio

$$(\text{Gross Profit} / \text{Net Sales}) \times 100$$

Operating Profit Ratio

$$\text{OP PROFIT (EBIT)} / \text{SALES} \times 100$$

Net Profit Ratio

$$(\text{Profit after tax} / \text{Net Sales}) \times 100$$

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Profitability Ratios



$\text{COGS} = \text{Opening Stock} + \text{Net Purchases} + \text{Direct Expenses} - \text{Closing Stock}$	$\text{Gross Profit} = \text{Sales} - \text{COGS}$
$\text{EBIT} = \text{Sales} - \text{all expenses except interest and tax}$	

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Profitability Ratios



Earnings Per Share

$$(\text{Net Profit after interest and Tax} - \text{Pref Dividend}) / \text{Number of Equity Shares}$$

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Profitability Ratios

Return on Equity

$$\frac{\text{PAT} - \text{Div} \times 100}{\text{Equity Shareholders Fund}}$$

 ((Net Profit after Taxes-Preference Dividend) /Equity Shareholders Funds)*100

Return on Investment /Return on Capital Employed

$$\frac{\text{Earnings before interest and Taxes}}{\text{Capital Employed}} \times 100$$

Handwritten notes for Return on Investment:

$$\frac{\text{EBIT}}{\text{Capital Employed}} \times 100$$

 Capital Employed $\rightarrow$ Debt + Equity
 Shareholders Funds

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Capital Employed

Debt +Equity Capital

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Solvency Ratios

Debt-Equity Ratio

Debt/Equity

Debt Capital Ratio

Debt/(Debt+equity)

Interest Coverage Ratio

$$\frac{\text{Earnings before interest and Taxes}}{\text{Interest Paid}}$$

Handwritten formula:
$$\frac{\text{EBIT}}{\text{Interest Paid}}$$

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Turnover Ratios

Debtors Turnover Ratio

Net Credit Sales/Average Debtors

Handwritten notes for Debtors Turnover Ratio:

$$\frac{\text{Total Sales} - \text{Sales Return}}{\text{Average Collection Period/Days Sales outstanding (DSO)}}$$

$$\frac{\text{Debtors P.P.} + \text{Debtors E.P.}}{2}$$

365/Debtors Turnover Ratio

These formulae can be calculated in weeks(54 weeks or 12 months also)

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Turnover Ratios

Creditors Turnover Ratio

Net Credit Purchases/Average Payables

Total Credit Purchases - Purchase Return
~~Average Payment period/Days Payables Outstanding (DPO)~~
 ~~$(\frac{365}{\text{Creditors Turnover Ratio}})$~~

365/Creditors Turnover Ratio

CTR

Cost of Good Sold= Opening Stock+ Net Purchases+ Direct Expenses-Closing Stock

Sales - COGS= GP

These formulae can be calculated in weeks(54 weeks or 12 months also)

Inventory Turnover Ratio=

COGS/Average Inventory at Cost

Inventory Turnover Days /Days Inventory Outstanding(DIO)
 365/Inventory Turnover Ratio

Cost of Good Sold=

Cost of Good Sold=

Opening Stock+ Net Purchases + Direct Expenses-Closing Stock

Sales- Gross Profit

Fixed Asset Turnover Ratio =

Revenue from operations/Net Fixed Assets

FA - Depreciation

Working Capital Turnover Ratio

$$= \frac{\text{COGS}}{\text{Net WC}}$$
Net WC = CA - CL
Gross WC = CA

Valuation Ratios

Price-to-earnings ratio (P/E ratio)

Market price per share /EPS

Compares a company's share price to its earnings per share. This ratio indicates whether a stock is expensive or cheap, and how much investors are willing to pay for each dollar of earnings.

Du pont Analysis (Three Step Analysis)

- 1 Profitability-Net Profit Margins → $\frac{\text{Net Profit}}{\text{Revenues}}$
- 2 Efficiency- Assets Turnovers → $\frac{\text{Revenues}}{\text{Average Total Assets}}$
- 3 Leverage- Equity Multiplier → $\frac{\text{Average Total Assets}}{\text{Average Shareholders Equity}}$

$$ROE = \text{Net Profit Margins} \times \text{Asset Turnover} \times \text{Equity Multiplier}$$

$$= \frac{\text{Net Profit}}{\text{Revenues}} \times \frac{\text{Revenues}}{\text{Avg Assets}} \times \frac{\text{Average Total Assets}}{\text{Average Shareholders Equity}}$$

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Commercial Banks

- Commercial banking (items fitted into the same categories as the non- financial firms)
- Firms in the financial industry could be expected to have high receivables (loans) and payables (deposits, other loans),
- a large ratio of assets to stockholders' equity (Dupont leverage), and little inventory.
 - Financial firms would also be expected to have low asset turnover ratios. They might also have a high gross profit percentage, i.e., net interest margin (interest revenue less interest income).
- From Exhibit 4, #7 has day's receivable of 3,208.07 days, a Dupont leverage ratio of 9.05, and an asset turnover ratio of 0.06. Thus, #7 is the commercial banking industry.

- Assigning the firms to the industry can be done using a four-step approach.
- First, organize the industries into several groups with broadly similar attributes.
- Second, determine the detailed financial characteristics of each of the broad groups.
- Third, identify the firms with the characteristics of those groups. Finally, match the firms to the specific industries making more detailed use of the financial information provided in the case. For example, the industries in the case can be organized into five groups consisting of:
 - (1) financial industry (commercial bank),
 - (2) retail industry (retail grocery stores),
 - (3) high technology (computer software, pharmaceutical preparations, and semiconductor manufacturer),
 - (4) service industry (mobile phone service provider), and
 - (5) capital intensive industry (commercial airline, integrated oil and gas, and liquor producer and distributor).

Retail Firms

- Retail firms would be expected to
 - have lots of inventory,
 - relatively low gross profit percentage,
 - and low returns on sales (less than 4%),
 - and high asset turnover.
- From Exhibit 2, firms #6 and #8 have a return on sales of 2.6% and 3.5%, respectively. Firm #6 has the lowest gross profit with 22.7%. From Exhibit 3, firm #8 has high inventories—21.5% of assets versus 2.4% for firm #6. From Exhibit 4, firm #8 has an asset turnover ratio of 2.31 versus 0.90 for firm #6. Thus, #8 is the retail grocery store industry.

High-Technology Industries



High Technology Industries

- Computer software
- R&D-based pharmaceutical manufacturer
- R&D-based semiconductor manufacturer

High technology industries would be expected to have

- high research and development intensity,
- and high return on sales (patent protection, brand value),
- And have low debt ratios because of the low collateral value of intangibles or growth opportunities.
- From Exhibit 2, firms #3, #4, and #5
- all have the highest gross profit margins and return on sales.

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Service Industry



•Service Industry

- Mobile phone service operator
- Mobile phone service operators would be expected to have
 - high receivables,
 - low inventories, and
 - high property, plant, and equipment.
- Company #1 has these characteristics, and so it is the mobile phone service provider.

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A pharmaceutical manufacturer would be expected to have the highest gross profit margin and high inventories, and a high day's inventory relative to the other two firms.

- From Exhibit 3, firm #4 has inventories of 7.6% of assets.

From Exhibit 4, firm #4 has day's inventory of 264 days

So, firm #4 is the pharmaceutical manufacturing industry. A software company would be expected to have proportionately lower inventories and plant, property, and equipment.

- From Exhibit 3, firm #5 has inventories of 1.3% of assets and plant, property, and equipment of 7.1% of assets. So, firm #5 is the software company. Firm #3,
- and #9, with plant, property, and equipment of 33.5% of assets, is the semiconductor.

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Capital-Intensive Industries



- Liquor producer and distributor
- Discount airline
- Large integrated oil and gas company
- A liquor company would have
 - high day's inventory (spirits must age with time),
 - and high selling, general administrative expenses because the product is heavily advertised and branded.
- From Exhibit 3, firm #2 has inventories of 21.6% of assets, and from Exhibit 4, firm #2 has a day's inventory of 342 days. Therefore,
- #2 is the liquor producer and distributor.
- Industries #6 and #9 are the remaining companies.
- An oil and gas company would be expected to have proportionately more inventories than an airline.
- An airline might be more capital intensive.
- From Exhibit 4, #6 and #9 have day's inventory of 12.5 days, and 16.7 days.
- From Exhibit 3, #6 and #9 have a ratio of PPE to assets of 68.3% and 67.7%, respectively. Airlines are often financed with debt.
- Firms #6 and #9 have total liabilities of 62% and 49% of assets, respectively.
- This suggests that firm #6 is the airline
- and firm #9 is the oil and gas firm.

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Company	Industry	Company
1	Mobile phone service operator	AT&T
2	Liquor producer and distributor	Brown-Forman
3	R&D-based semiconductor manufacturer	Intel
4	R&D-based pharmaceutical manufacturer	Eli Lilly
5	Computer software company	Microsoft
6	Discount airline	Southwest Airlines
7	Commercial bank (items fitted into the same categories as the non-financial firms)	Wells Fargo
8	Retail grocery company	WalMart
9	Large integrated (exploration and production) oil and gas company	ExxonMobil



Thank You!



Managerial Accounting

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MBACCZG502, Managerial Accounting Lecture No.

Ratio Analysis – From Company's Perspective

Abhishek Sinha

Learning Objective

- At the end of the session you should be able to calculate various financial ratios
- Interpret and understand the significance of the ratios to an organization
- Appreciate the different approaches used to calculate the ratios
- Understand the difference between “drunkard” and “illumination” approach of ratio analysis

Terms to know before we start

Accounting Statements	Terms
Income Statement	Sales, Cost of Goods Sold, SG&A expenses, Depreciation, EBITDA, EBIT, Gross Profit, Net Profit, Interest Expenses, Interest Income, Other Income, Extraordinary Expenses, Income from Discontinued Business, Adjusted Net Profit, Dividends, Retained Profits
Balance Sheet	Shareholders Fund, Long Term Liability, Capital Employed, Current Liability, Current Asset, Quick Assets, Fixed Assets, Fictitious Assets, Total Assets, Shares Outstanding, Capital Expenditure, Enterprise Value
Cash Flow	Cash flow from operating activities, cash flow from investing activity, cash flow from financing activity, Free Cash Flow

Introduction

- A method by which relationship of items or groups of items in the financial statements are computed, and presented
- Is used to interpret the financial statements so that the strengths and weaknesses of a firm, its historical performance and current financial condition can be determined

Caveat for students

- A ratio is not an end in itself. They are only a means to get to know the financial position of an enterprise
- The process of selecting inputs is subjective
- May need to be supplemented by qualitative factors to get a complete picture
- Computing ratios only reveals the relationship in a more meaningful way so as to enable us to draw conclusions there from

Ratios – A Snap Shot



$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Net Sales}}$$



Net Sales

- Volume
- Price per unit

Cost of Goods Sold

- Raw Material Consumed
- Direct Expenses

$$\text{Net Profit Ratio} = \frac{\text{Net Profit}}{\text{Net Sales}}$$



Net Sales

- Volume
- Price per unit

Net Profit

- Total expenses

Operating Profit Ratio = Operating Profit/ Net Sales

Net Sales

- Volume
- Price per unit

Operating Expenses

- Raw Material Consumed
- Direct Expenses
- SG&A

Cost Of Goods Sold

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Return on Shareholders Fund=

Net Profit
/ Shareholders Fund

Net Profit

Shareholders Fund

- Equity Share
- Preference Shares
- Minority Interest
- Reserves and Surplus

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Return on Capital Employed=

NOPAT/ Total Capital Employed

Net Operating
Profit after tax

Shareholders Funds

- Equity Share
- Preference Shares
- Minority Interest
- Reserves and Surplus
- Long Term Liabilities

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Current Ratio=

Current Asset/ Current Liability

Current Asset

- Debtors
- Cash and Cash Equivalents
- Prepaid Exp
- Receivable Income
- Inventory

Current Liability

- Creditors
- O/s Expenses
- Prereceived income
- Creditors
- Provisions for Tax

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Current Ratio=

Current Asset/ Current Liability

Current Asset

- Debtors
- Cash and Cash Equivalents
- Prepaid Exp.
- Receivable Income
- Inventory

Current Liability

- Creditors
- O/s Expenses
- Prereceived income
- Creditors
- Provisions for Tax

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Quick Ratio=

Quick Asset/ Current Liability

Quick Asset

- Debtors
- Cash and Cash Equivalents
- Receivable Income

Current Liability

- Creditors
- O/s Expenses
- Prereceived income
- Creditors
- Provisions for Tax

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Price / Earning Ratio=

Market Price/ EPS

Market Price

- Trading Price of listed stock

Earning Per Share

- Net Profit divided by no. of shares outstanding

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Price / Book Value=

Market Price/ Book Value

Market Price

- Trading Price of listed stock

Book Value Per Share

- Total Assets Less Fictitious Asset
- Less : Current Liability
- Less: Long Term Loan

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EV / Sales

Enterprise Value

- Market Capitalization
- Long Term Debt
- Minority Interest
- Less: Cash and Cash Equivalents

Net Sales

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EV / EBITDA

Enterprise Value

- Market Capitalization
- Long Term Debt
- Minority Interest
- Less: Cash and Cash Equivalents

EBITDA

- EBIT and Depreciation and Amortization

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Debtors Turnover

Net Sales

Debtors

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Debtors Days

- Debtors days estimates the average time it takes a business to collect its debts with trade suppliers. As an approximation of the amount spent with trade creditors, the convention is to use cost of sales in the formula which is as follows
- Debtors Days = $365 / (A/R \text{ Turnover})$

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Creditors Turnover



- Creditor days estimates the average time it takes a business to settle its debts with trade suppliers. As an approximation of the amount spent with trade creditors, the convention is to use cost of sales in the formula which is as follows

$$\text{Creditor days} = \frac{\text{Trade payables}}{\text{Cost of sales}} \times 365$$

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Du Pont Ratio

Return on Equity = Net Profit / Shareholders Fund

Net Profit Margin = Net Profit / Net Sales

Asset Turnover = Net Sales / Assets

Equity Multiplier = Asset / Shareholders Fund

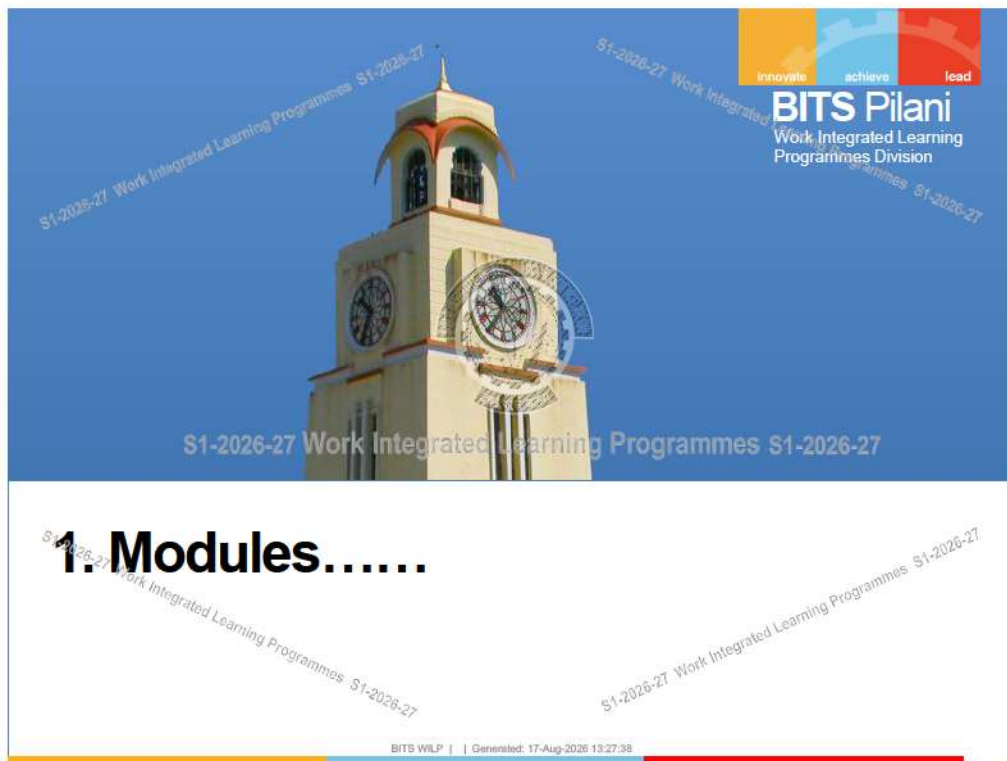
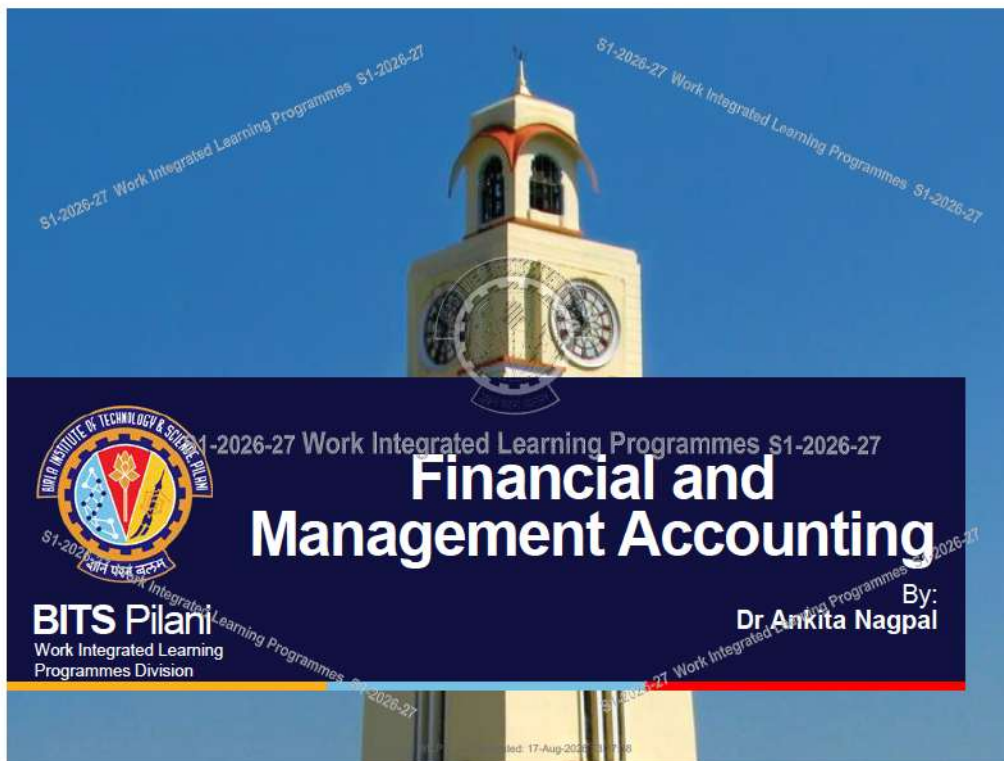
Return on Equity = Net Profit Margin * Asset Turnover * Equity Multiplier

Return on Equity = f(Profitability, Asset Utilization and Financial Leverage)

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Thank You!



1. Modules.....

Cash flow statement

The balance sheet provides information about an enterprise's assets and how those assets have been financed. But it **does not explain the changes during a period in assets, liabilities, and equity resulting from the enterprise's activities.**

The statement of profit and loss provides information about an enterprise's financial performance. However, **since earnings are measured by accrual accounting, they don't show the cash generated during the year.**

A Cash Flow Statement is a statement which shows the **changes in financial position of a firm on cash basis.**

All the cash inflows during a particular period are shown in the statement from which the cash outflows are deducted for the same period.

It thus shows the **sources of cash and the applications(uses) of cash.** As a result it also shows the net cash in hand at the end of the year

Categorization of cash flows



The cash flows are categorized into three main heads as those related to the following:

- ① Operating activities → Day to day operation
- ② Investing activities ✓
- ③ Financing activities ✓

↳ generating the cash for the business

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Financial and Management Accounting - 5

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Investing activities



Investing activities include **actual investments** made in projects, asset acquisition etc.. Cash flow from investing activities includes money spent on **acquisition of assets and the money received from the sale of assets**. Investments and their redemption such as money spent on **purchasing shares of other companies, loans given to others etc** are also included here.

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Financial and Management Accounting - 7

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Operating activities



Operating activities include all the activities related to the **actual business operations**. The actual manufacture, sales and delivery of the product in which the organization deals are operating activities. The cash flows from operating activities could include the inflows and outflows related to the above activities.

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Financing activities



Financing activities include activities related to obtaining **finance for the business in terms of shares, debentures or loans.. Dividend paid to shareholders as well as interests paid on loans** become the cash outflows from financing activities.

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Financial and Management Accounting - 8

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Accounting Equation



There are two ways of computing and reporting net cash flow from operating activities:

- The direct method
- The indirect method



Statement of Cash Flows



Net Profit
Add/Less: Income /Expenses from Operating Activities
Add/Less: Income /Expenses from Long term investing Activities
Add/Less: Income /Expenses from Financing Activities

Statement of Cash Flows

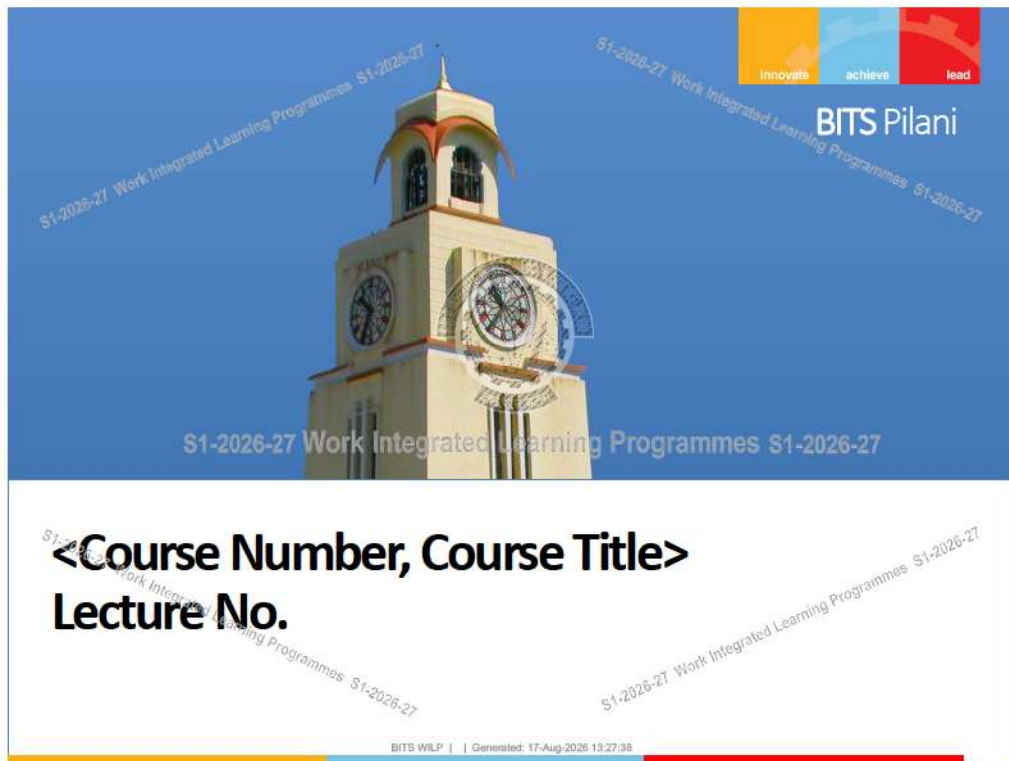


Net Decrease/Increase in Cash
Add: Cash at the beginning of the year

Cash at the end of the year



Thank You!



Cash Flow

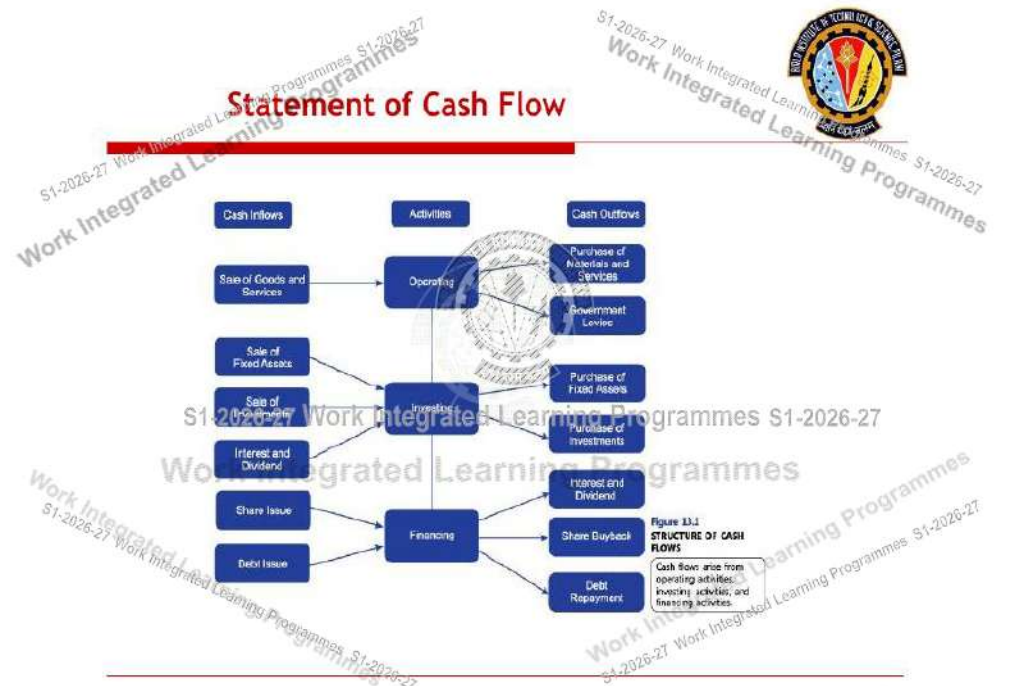
Cash Flow Classification

Category	Inflows	Outflows
Operating activities: Producing and delivering goods and services.	- Cash sales. - Unearned revenue. - Collection of trade receivables.	- Purchase of materials and services. - Payment of salaries and benefits. - Payment of taxes and duties.
Investing activities: Acquiring and disposing of property, plant and equipment and investments.	- Sale of property, plant and equipment. - Sale of investments. - Sale of businesses. - Receipt of interest and dividend.	- Purchase of property, plant and equipment. - Purchase of investments. - Acquisition of businesses.
Financing activities: Obtaining and returning funds.	- Issue of shares. - Issue of bonds. - Proceeds from loans.	- Buyback of shares. - Repayment of bonds. - Repayment of loans. - Payment of interest and dividend.

Non-cash investing and financing activities: Transactions that affect assets or liabilities but do not result in cash inflows or outflows.

Examples:

- Converting debt into equity.
- Converting preference shares into equity.
- Purchasing a building by incurring a mortgage to the seller.
- Obtaining an asset on a finance lease.
- Exchanging a non-cash asset for another non-cash asset.



Classify each as an operating activity, investing activity, financing activity, non-cash activity, or cash management activity.



- (a) Deposited cash in the current account. **(Operating, Investment and Financing)**
- (b) Paid bonus to employees.
- (c) Acquired equipment on credit.
- (d) Paid cash on maturity of bills payable for purchases of goods.
- (e) Paid income tax.
- (f) Paid cash to settle a suit for trademark infringement.
- (g) Purchased a 60-day certificate of deposit.
- (h) Issued equity shares on conversion of debentures.
- (i) Acquired a machine on hire purchase.
- (j) Issued equity shares at par Rs.200 crore.
- (k) Wrote off a receivable when a customer became insolvent.
- (l) Issued equity shares in exchange for preference shares.
- (m) Received cash on the maturity of bills receivable.

3

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Classify transactions into operating activity, investing activity, financing activity, non-cash activity, or cash management activity.



Sohan Company entered into the following transactions.

- (a) Paid suppliers.
- (b) Received dividend from an associate.
- (c) Sold investments at a gain.
- (d) Purchased copyrights for cash.
- (e) Issued debentures in exchange for equipment.
- (f) Paid interest on bank overdraft.
- (g) Converted debentures into equity shares.
- (h) Received payments from customers.
- (i) Purchased investments for cash.
- (j) Purchased a 60-day certificate of deposit for cash.
- (k) Paid advance to a supplier of equipment.
- (l) Sold a 90-day Treasury bill.
- (m) Collected instalment payments for the sale of an old plant.
- (n) Earned a net profit.

4

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- (o) Disposed of equipment at a gain.
- (p) Entered into a finance lease for a new machine.
- (q) Sold equipment for cash at carrying amount.
- (r) Issued convertible debentures for cash.
- (s) Paid festival bonus to employees.
- (t) Collected insurance proceeds from a patent infringement suit.
- (u) Paid instalments for the purchase of inventories.

Required

1. Classify each as an operating activity, investing activity, financing activity, non-cash activity, or cash management activity.

5

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Classify transactions into operating activity, investing activity, financing activity, non-cash activity, or cash management activity.



Manu Company entered into the following transactions.

- (a) Collected instalment payments for the sale of inventories.
- (b) Paid income tax.
- (c) Entered into a hire purchase for a new plant.
- (d) Purchased goods for cash.
- (e) Issued debentures.
- (f) Paid a dividend.
- (g) Issued commercial paper with a maturity of 60 days.
- (h) Received interest on investment in Treasury bill.
- (i) Exchanged equipment for shares.
- (j) Made a two-year loan to a subsidiary.
- (k) Paid principal amounts of price for instalment purchase of the plant.
- (l) Converted non-current loan into equity shares.

6

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- (m) Transferred cash to a six-month deposit account.
- (n) Disposed of equipment at a loss.
- (o) Exchanged a building for a plant.
- (p) Repaid bank overdraft.
- (q) Retired equipment from use.
- (r) Sold goods for cash.
- (s) Issued shares at par.
- (t) Disposed of patents at a gain.
- (u) Made a down payment for the purchase of equipment.

Q: Classify each as an operating activity, investing activity, financing activity, non-cash activity, or cash management activity.

Practice

Haveli Company had the following investing activities during the year ended March 31, 20X7:

- (a) Purchased investments, Rs. 6500
- (b) Sold investments, Rs.3,800 (loss, Rs.400).
- (c) Purchased a plant, Rs.15,800.
- (d) Constructed a building for own use, Rs.7,800 (including capitalized interest, Rs.150).
- (e) Sold a plant, Rs.5,200 (cost, Rs.9,500; carrying amount, Rs. 4,300).

Required

Compute net cash flow from investing activities.

Problem 13.10

Cash Flows from Operating Activities ***

Gopal Dairy Company's statement of profit and loss for the year ended September 30, 20X7 is as follows:

GOPAL DAIRY COMPANY Statement of Profit and Loss For the year ended September 30, 20X7	
Sales	₹27,100
Interest income	300
Gain on sale of investments	100
Total income	₹27,500
Expenses	
Cost of goods sold	18,800
Depreciation expense	4,800
Finance costs	500
Selling and administrative expenses	1,900
Loss on sale of equipment	200
Total expenses	₹26,100
Profit before tax	₹1,400
Tax expense	300
Profit for the period	₹1,100

PBT + Non-Cash Expenses + Non-Operating Expenses - Non-Operating Incomes

Increase in CA will lead to Decrease in Operating Cash Flow and Vice versa.
Increase in CL will increase in Operating Cash Flow and vice versa

During the period, inventories decreased by '900, trade receivables (gross) increased by '1,200, trade payables decreased by '700, and income tax payable increased by '300. Selling and administrative expenses include bad debt expense of '400. Trade receivables in the amount of '200 were written off during the period.

Required

1. Present the cash flows from operating activities using the direct method.
2. Present the cash flows from operating activities using the indirect method.



Cash Flow

Cash Flow Classification

Category	Inflows	Outflows
Operating activities: Producing and delivering goods and services.	• Cash sales. • Unearned revenue. • Collection of trade receivables.	• Purchase of materials and services. • Payment of salaries and benefits. • Payment of taxes and duties.
Investing activities: Acquiring and disposing of property, plant and equipment, and investments.	• Sale of property, plant and equipment. • Sale of investments. • Sale of businesses. • Receipt of interest and dividend.	• Purchase of property, plant and equipment. • Purchase of investments. • Acquisition of businesses.
Financing activities: Obtaining and returning funds.	• Issue of shares. • Issue of bonds. • Proceeds from loans.	• Buyback of shares. • Repayment of bonds. • Repayment of loans. • Payment of interest and dividend.
Non-cash investing and financing activities: Transactions that affect assets or liabilities but do not result in cash inflows or outflows.		
Examples:		
• Converting debt into equity. • Converting preference shares into equity. • Purchasing a building by incurring a mortgage to the seller. • Obtaining an asset on a finance lease. • Exchanging a non-cash asset for another non-cash asset.		

Cash Flow

Cash Flow Classification

Category	Operating Activities	Investing Activities	Financing Activities
Operating activities: Producing and delivering goods and services.	• Cash sales • Cash receipts from interest • Collection of trade receivables	• Sale of property, plant and equipment • Sale of investments • Sale of securities • Proceeds from sale of subsidiary	• Issuance of shares • Issuance of bonds • Proceeds from loans
Investing activities: Acquiring and disposing of property, plant and equipment and investments.	• Purchase of property, plant and equipment • Purchase of investments • Purchase of securities • Purchase of subsidiary	• Sale of property, plant and equipment • Sale of investments • Sale of securities • Proceeds from sale of subsidiary	• Issuance of shares • Issuance of bonds • Proceeds from loans
Financing activities: Raising and repaying funds.	• Issuance of shares • Issuance of bonds • Proceeds from loans	• Sale of property, plant and equipment • Sale of investments • Sale of securities • Proceeds from sale of subsidiary	• Issuance of shares • Issuance of bonds • Proceeds from loans

Classify each as an operating activity, investing activity, financing activity, non-cash activity, or cash management activity.

- Deposited cash in the current account. (Operating, Investment and Financing)
- Paid bonus to employees.
- Acquired equipment on credit.
- Paid cash on maturity of bills payable for purchases of goods.
- Paid income tax.
- Paid cash to settle a suit for trademark infringement.
- Purchased a 60-day certificate of deposit.
- Issued equity shares on conversion of debentures.
- Acquired a machine on hire purchase.
- Issued equity shares at par Rs. 200 crore.
- Wrote off a receivable when a customer became insolvent.
- Issued equity shares in exchange for preference shares.
- Received cash on the maturity of bills receivable.

Statement of Cash Flow



Classify transactions into operating activity, investing activity, financing activity, non-cash activity, or cash management activity.

- Sohan Company entered into the following transactions.
- Paid suppliers.
 - Received dividend from an associate.
 - Sold investments at a gain.
 - Purchased copyrights for cash.
 - Issued debentures in exchange for equipment.
 - Paid interest on bank overdraft.
 - Converted debentures into equity shares.
 - Received payments from customers.
 - Purchased investments for cash.
 - Purchased a 60-day certificate of deposit for cash.
 - Paid advance to a supplier of equipment.
 - Sold a 90-day Treasury bill.
 - Collected instalment payments for the sale of an old plant.
 - Earned a net profit.

Classify transactions into operating activity, investing activity, financing activity, non-cash activity, or cash management activity.

- Manu Company entered into the following transactions.
- Collected instalment payments for the sale of inventories.
 - Paid income tax.
 - Entered into a hire purchase for a new plant.
 - Purchased goods for cash.
 - Issued debentures.
 - Paid a dividend.
 - Issued commercial paper with a maturity of 60 days.
 - Received interest on investment in Treasury bill.
 - Exchanged equipment for shares.
 - Made a two-year loan to a subsidiary.
 - Paid principal amounts of price for instalment purchase of the plant.
 - Converted non-current loan into equity shares.

- Disposed of equipment at a gain.
- Entered into a finance lease for a new machine.
- Sold equipment for cash at carrying amount.
- Issued convertible debentures for cash.
- Paid festival bonus to employees.
- Collected insurance proceeds from a patent infringement suit.
- Paid instalments for the purchase of inventories.

Required

- Classify each as an operating activity, investing activity, financing activity, non-cash activity, or cash management activity.

- Transferred cash to a six-month deposit account.
- Disposed of equipment at a loss.
- Exchanged a building for a plant.
- Repaid bank overdraft.
- Retired equipment from use.
- Sold goods for cash.
- Issued shares at par.
- Disposed of patents at a gain.
- Made a down payment for the purchase of equipment.

Q: Classify each as an operating activity, investing activity, financing activity, non-cash activity, or cash management activity.

- Haveli Company had the following investing activities during the year ended March 31, 20X7:
- Purchased investments, Rs. 6500.
 - Sold investments, Rs. 3,800 (loss, Rs. 400).
 - Purchased a plant, Rs. 15,800.
 - Constructed a building for own use, Rs. 7,800 (including capitalized interest, Rs. 150).
 - Sold a plant, Rs. 5,200 (cost, Rs. 9,500; carrying amount, Rs. 4,300).

Required

Compute net cash flow from investing activities.

Problem 13.38
Cash flow from operating activities

Suppose that the company's statement of profit and loss for the year ended September 30, 20X7 is as follows:

GENERAL COMPANY	
Statement of Profit and Loss for the year ended September 30, 20X7	
	Rs.
Sales	100,000
Cost of sales of finished goods	(60,000)
Gross profit	40,000
Operating expenses:	
Cost of goods sold	(10,000)
Depreciation expense	(5,000)
Interest expense	(2,000)
Selling and administrative expenses	(10,000)
Loss on sale of equipment	(1,000)
Gain on sale of land	3,000
Income tax expense	(2,000)
Net income	13,000

During the period, inventories decreased by 100, trade receivables (net) increased by 1,200, trade payables decreased by 700, and income tax payable increased by 300. Selling and administrative expenses include bad debt expense of 400. Trade receivables in the amount of 200 were written off during the period.

Required

- Present the cash flows from operating activities using the direct method.
- Present the cash flows from operating activities using the indirect method.

S1-2026-27 Work Integrated Learning Programmes S1-2026-27



BITS Pilani
Work Integrated Learning
Programmes Division

S1-2026-27 Work Integrated Learning Programmes S1-2026-27
**Financial and
Management Accounting**

By: Dr Ankita Nagpal



1. Modules.....

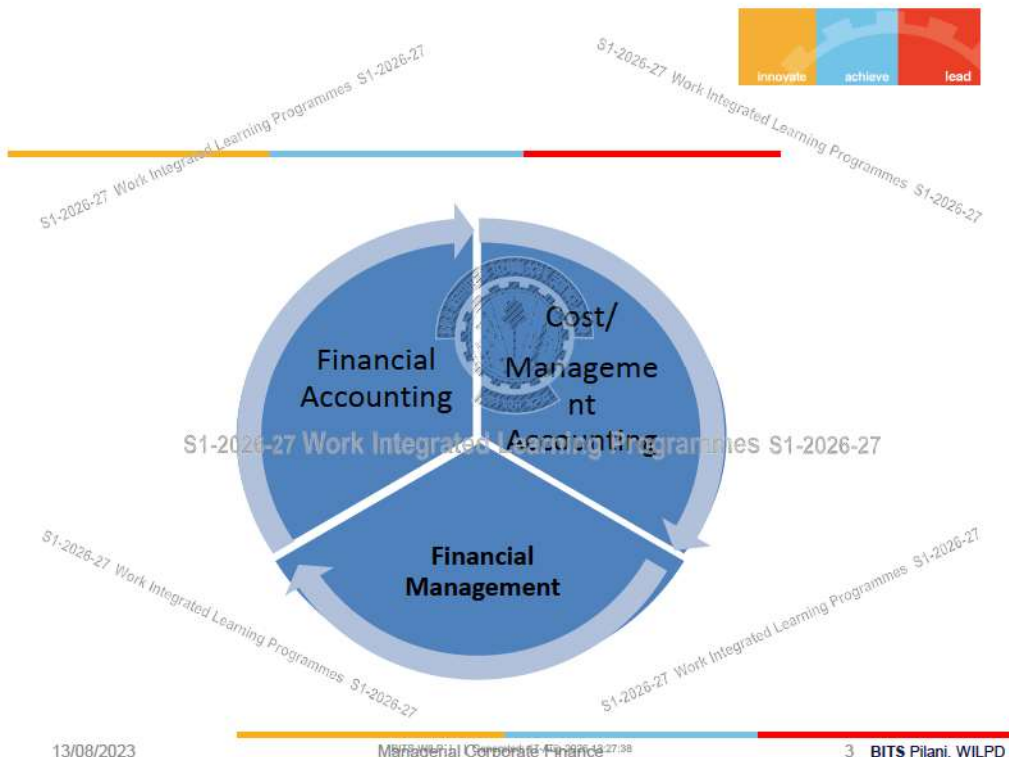
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Management accounting

Management accounting is the classifying, recording and appropriate allocation of expenditure for determining the costs of products and services.

S1-2026-27 Work Integrated Learning Programmes S1-2026-27

It is aimed at producing and communicating information that is relevant to managerial decision making.



13/08/2023

Managerial Corporate Finance

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Categories of Costs

TRACEABILITY
DIRECT COSTS
INDIRECT COST

VARIABILITY/BEHAVIOUR
FIXED COST
VARIABLE COST
SEMI VARIABLE

Manufacturing
Administrative Exp.
Selling & Distribution
Salaries for white collar
Rent for factory
depends upon the
per unit production
20% → variable
80% → Fixed
Selling Exp.
Salaries

Categories of Costs

RELEVANCE TO DECISION MAKING

RELEVANT COSTS/AVOIDABLE COSTS
OPPORTUNITY COSTS

IRRELEVANT COSTS/UNAVOIDABLE COSTS
SUNK COSTS

TRACEABILITY

• Direct Costs (Prime Costs)

- Costs that are directly traceable to a product
- Direct Material ✓ *Raw Material cost*
- Direct Labour → *wages.*
- Direct Expenses

↳ Maintenance of Machinery
↳ Electricity Bill of Factory
↳ Cost

Direct materials

Direct materials are those materials that become part of the finished product. They are tracked on a **per-unit basis** when it is feasible and economically worthwhile to do so.

A significant, integral part of the finished product.

Direct labour

Direct labor refers to the work of employees that can be clearly and directly traced to a specific job, task, or unit of production.

Direct Expenses

• Any other costs (not material or labor) that are directly attributable to a specific product, job, or project.

• They do NOT become part of the product, but they are necessary to produce it.

Examples:

- Royalties paid per unit produced
- Special tools or molds made for a specific job
- Hire of a machine used only for a particular job
- Design costs for a custom project

Categories of Costs

Prime Cost

Direct Material + Direct Labour + Direct Expenses

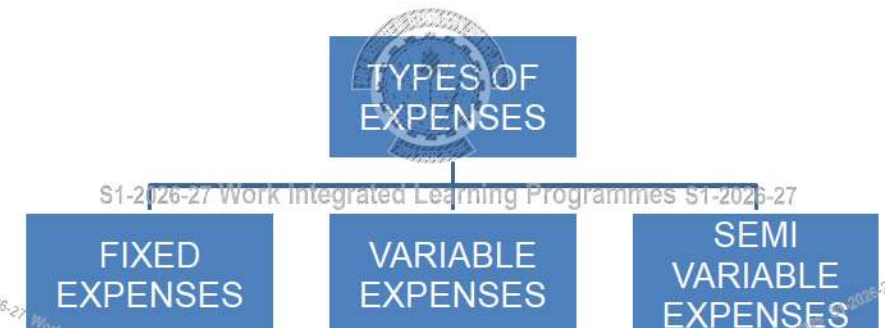
OverHeads = Ind mat + ind lab + IE

Categories of Costs

Indirect Costs(Overhead)

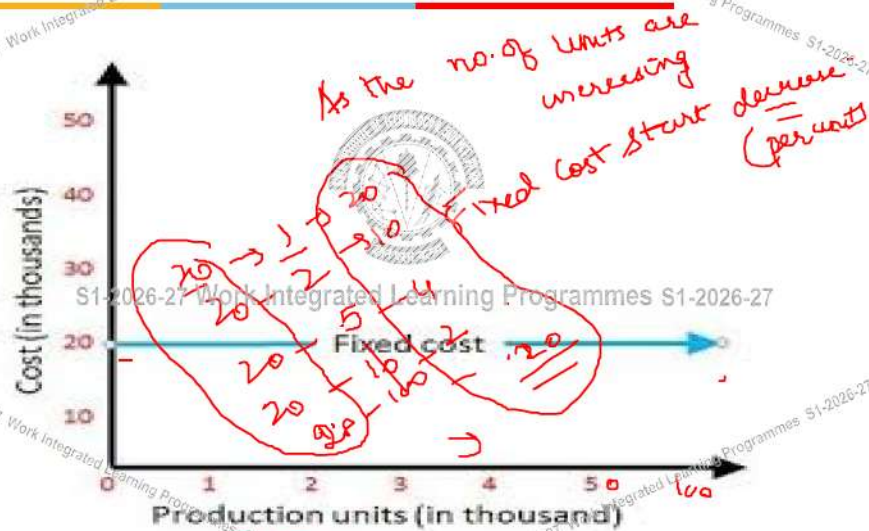
- Manufacturing Overheads
- Administrative Overheads
- Selling and Distribution overheads

VARIABILITY/BEHAVIOUR



Categories of Costs

- Fixed Costs: The total cost remains the same irrespective of units produced
- Rent of a building, salary of warehouse keeper, salary of factory manager



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Managerial Corporate Finance

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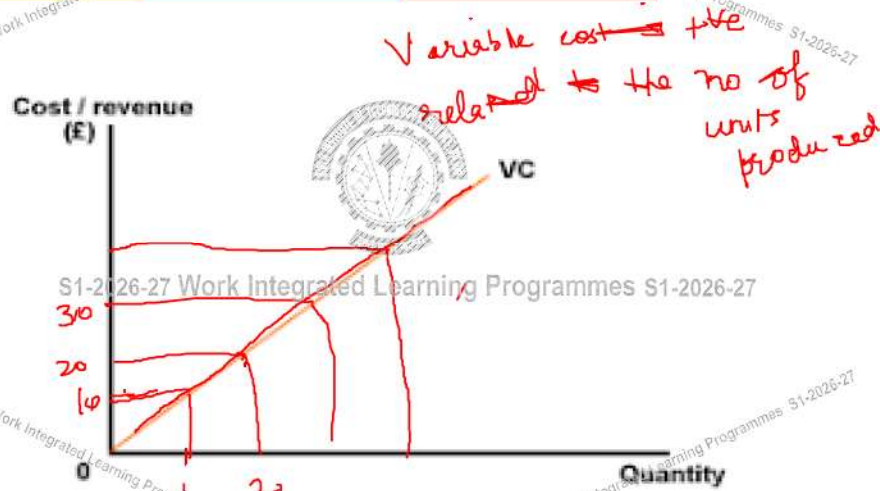
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Categories of Costs

- Variable Costs: The cost per unit remains the same, though the total cost changes linearly with output
- Direct material, direct labour are always fully variable



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Managerial Corporate Finance

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Categories of Costs

- Direct material, direct labour are always fully variable
- Eg: cost of refill of a gel pen is Rs 20

	Cost per refill	Total cost
1	20	20
2	20	40
3	20	60
4	20	80

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Managerial Corporate Finance

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Categories of Costs

- Eg, cost of milk(250 ml) in 1 large cup of coffee in coffee store

	Cost per cup	Total cost
1	15	15
2	15	30
3	15	45
4	15	60

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Managerial Corporate Finance

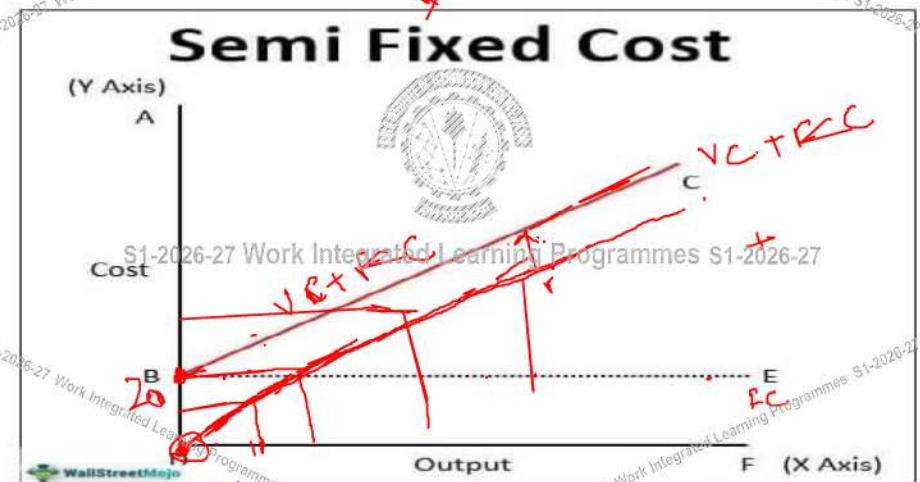
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Marginal cost/Variable cost

The change in total cost for an increase or decrease in one unit of output.

$$TC = FC + VC$$

Semi Fixed Cost



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Skoda Octavia, a plant of New United Motor Manufacturing Ltd (NUMML), a joint venture of General Motors and Toyota, assembles two types of cars—Corolla and Geo Prisms.

Separate assembly lines are used for each car.

Classify the following costs into

Direct (D) or Indirect (I), and Fixed (F) or Variable (V) for each assembly line

D.V.

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Cost Item

- Cost of Tires used for Geo Prisms
- (DV)
- Salary of Public Relations manager for NUMML Plant
- (IF)
- Awards dinner for Corolla suppliers
- (DV)
- Salary to engineer who monitors design changes on Geo Prism
- (DF)

4/30/2018

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Cost Item

- Cost of Tires used for Geo Prisms
- Salary of Public Relations manager for NUMML Plant
- Awards dinner for Corolla suppliers
- Salary to engineer who monitors design changes on Geo Prism
- Freight Costs of Corolla engines shipped from Toyota City, Japan to Fremont, California
- Electricity costs for NUMML plant
- Wages paid to temporary assembly line workers hired in periods of high production (paid on hourly basis)
- Annual fire insurance policy for NUMML Plant

4/30/2018

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Cost Item

- Freight Costs of Corolla engines shipped from Toyota City, Japan to Fremont, California
- (D F)
- Electricity costs for NUMML plant
- (H V) — Fixed → 20%
- Wages paid to temporary assembly line workers hired in periods of high production (paid on hourly basis):
- (DV)
- Annual fire insurance policy for NUMML Plant:
- (IF)

4/30/2018

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Categories of Costs

RELEVANCE TO DECISION MAKING

RELEVANT COSTS/ OPPORTUNITY COSTS

IRRELEVANT COSTS

SUNK COSTS

An amount of Rs 40,000 was spent on the purchase of a software for employees in a company. However, in a few months the company realises that a newer and more efficient version of the software is available in the market which can improve productivity by 50% pa. Hence the company decided to go in for a new updated software available in the market costing Rs 1,00,000.

What is the sunk cost for the company?

Ans: 40,000

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Managerial Corporate Finance

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4/30/2018

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▶ A company purchased 1,000 units of a chemical at Rs 500 per unit. However, after purchase, it realized that the quality of chemical was not suitable for its production purposes and has no resale value.

▶ A buyer B is willing to buy this chemical at Rs 700 per unit if some conversion work is undertaken on this chemical. The cost of conversion of the chemical is Rs 700 per unit.

▶ Should the company undertake the conversion?-

▶ Yes

▶ What is the sunk cost for the co?

▶ $500 \times 1000 = 500,000$

▶ A company purchased 1,000 units of a chemical at Rs 500 per unit. However, after purchase, it realized that the quality of chemical was not suitable for its production purposes.

▶ A buyer B is willing to buy this chemical at Rs 700 per unit if some conversion work is undertaken on this chemical. The cost of conversion of the chemical is Rs 700 per unit.

▶ Another buyer C was willing to buy the chemical in its existing form. What should be the minimum price that C should be willing to pay for the company to sell the chemical to him?

▶ Zero

▶ The chemical is dangerous and it isn't feasible to store it for long. Disposing the chemical will cost Rs 200 per unit.

▶ Now, what should be the minimum price that C should be willing to pay for the company to sell the chemical to him?

▶ -200

4/30/2018

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4/30/2018

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- ▶ A company purchased 1,000 units of a chemical at Rs 500 per unit. However, after purchase, it realized that the quality of chemical was not suitable for its production purposes.
- ▶ Another buyer C was willing to buy the chemical in its existing form. What should be the minimum price that C should be willing to pay for the company to sell the chemical to him?
- ▶ **Zero**
- ▶ The chemical is dangerous and it isn't feasible to store it for long. Disposing the chemical will cost Rs 200 per unit.
- ▶ Now, what should be the minimum price that C should be willing to pay for the company to sell the chemical to him?
- ▶ **-200**

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Inventory valuation methods



Inventory valuation methods are used to determine the value of the inventory on hand, which ultimately affects the cost of goods sold (COGS) and, consequently, a company's profit and taxes. Most common methods used for inventory valuation:

FIFO

LIFO

WAC-Weighted Average

Handwritten notes:
Inventory / stock → COGS
↓ profitability

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- What is the cost per ticket to a professional cricket league team that offers 50 free tickets to an orphanage?
- Does it matter for what game the tickets are offered?
- Why would it probably cost the cricket league team more to give the tickets to college students than to poor orphans?

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Inventory valuation methods



FIFO (First-In, First-Out)

FIFO assumes that the oldest inventory items are sold first. Therefore, the items purchased first are the first ones used in the production or sold to customers.

Impact on Valuation:

In a period of **rising prices**, FIFO results in **lower COGS and higher inventory values**, which leads to **higher profits and higher taxes** because the older, cheaper inventory is sold first.

Best for:

Companies where inventory items are **perishable or have a shelf life**, as this method ensures older items are used first.

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Inventory valuation methods



LIFO (Last-In, First-Out)

LIFO assumes that the most recent inventory items are sold first. The latest purchases are considered the first goods to be sold.

Impact on Valuation:

In periods of rising prices, LIFO results in higher COGS and lower ending inventory, leading to lower profits and lower taxes. This is because the latest (and typically more expensive) inventory is sold first.

Best for:

Companies in industries where prices are rising rapidly and they want to reduce tax liabilities

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Budgeting



Segregation of costs helps in planning and preparing budgets for the forthcoming financial year

Cash Budget

Master Budget

Zero Base Budget

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Inventory valuation methods



Weighted Average Cost (WAC)

The weighted average cost method assigns a single average cost to each unit of inventory, calculated by dividing the total cost of goods available for sale by the number of units available. Every time new inventory is purchased, a new average cost is calculated.

Impact on Valuation:

The cost of goods sold and ending inventory are based on this average cost, leading to smoother, less volatile valuations over time. It is often used when inventory items are indistinguishable or when inventory turnover is high.

Best for: Companies dealing with large quantities of identical or similar items (e.g., bulk commodities).

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Cash Budget



A cash budget is a financial plan that estimates the cash inflows and outflows of a business over a specific period of time, typically monthly, quarterly, or annually. It helps businesses manage their cash flow by ensuring that they have enough liquidity to meet their short-term obligations and avoid cash shortages.

Key Elements of a Cash Budget:

Cash Inflows: This includes all the expected sources of cash, such as sales revenue, loans, capital contributions, or any other sources of incoming funds.

Cash Outflows: This includes all anticipated cash payments, such as operating expenses (e.g., rent, utilities, payroll), debt repayments, or purchases of inventory.

Net Cash Flow: The difference between the cash inflows and outflows during a period. A positive net cash flow indicates that the company has enough cash, while a negative net cash flow indicates a shortfall.

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•Advantages:

- Helps businesses **ensure liquidity** by predicting cash needs and shortages.
- Assists in managing **short-term financial obligations** and planning for future expenditures.
- Guides financial decision-making** regarding borrowing, investing, or managing working capital.

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Master Budget

A **master budget** is a comprehensive financial plan that consolidates all individual budgets within a business. It includes various budgets such as sales, production, operating expenses, cash flow, and capital expenditures. The master budget provides an overall financial picture of the company, helping management to plan and coordinate activities across departments.

Key Components of a Master Budget:

Sales Budget: Forecasts expected sales revenue based on anticipated units sold and the expected price.

Production Budget: Estimations of the quantity of goods to be produced, based on the sales forecast and inventory levels.

Direct Materials Budget: Estimates the cost of raw materials required for production.

Direct Labor Budget: Estimates labor costs based on the number of workers needed and their wage rates.

Overhead Budget: Estimates costs associated with operating the business, including rent, utilities, and depreciation.

Cash Budget:

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Zero base Budget

Zero-Base Budgeting (ZBB) is a budgeting method where every expense must be justified for each new period, starting from a "zero base." Unlike traditional budgeting, where previous budgets or historical data are used as a starting point, ZBB requires managers to build their budgets from scratch each time, justifying every cost and resource allocation, regardless of past expenditures.

Key Features of Zero-Base Budgeting:

Start from zero: Every department begins with a budget of zero and must justify all expenses as if it were the first time the budget is being prepared.

Justification of Expenses: Every line item or expense must be justified based on its importance and necessity. This can include justifying the benefits of continuing a project, hiring a new employee, or purchasing new equipment.

Prioritization: Resources are allocated based on the most important and value-adding activities, rather than relying on historical budget allocations.

•Advantages:

- Comprehensive planning:** Provides a holistic view of the company's financial activities.
- Coordination:** Helps align the objectives of different departments (e.g., sales, production, finance) toward a common organizational goal.
- Financial control:** Allows for performance monitoring by comparing actual results with budgeted figures.
- Decision-making:** Helps management make informed decisions regarding investments, financing, and operations.

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Special Orders

Deciding whether to accept a special order is a short-run decision. Management must decide:

• What sales price is appropriate for one-time customers?

• Does the company have excess capacity?

• Can it produce additional units with its existing resources?



Special Order - Example

Imagine that Sunset Airlines, a major regional airlines, has been asked by a large corporation to provide 150 seats to executives attending a conference.

- The corporation offers \$125 per ticket.
- The normal fare is \$275.
- The ticket is only good for one day for any of 5 flights.
- The aircraft carries 180 passengers (or $5 \times 180 = 900$ for one day).
- The normal load is 700 per day.

Sunset Airlines

Since Sunset has **excess capacity**, should they accept the special order?

Excess Capacity

Excess Capacity

Excess Capacity

Special orders are almost never accepted if a company does not have excess capacity, because current customers may have to be turned away.

Even when excess capacity exists and a special order is profitable, companies must consider **qualitative factors**.

Special Order - Example

What are the OPTIONS?

1. Sell the special order tickets at \$125.
2. Sell only the \$275 market price tickets.
3. Sell the special order tickets at another price.

Sunset Airlines

Full Cost Analysis

What is the objective? To maximize income in the short run without reducing income in the long run. Here are the full cost data.

	Per Passenger	Per Round-trip
Cost of meals	\$ 6.50	\$ 1,170
Cost of fuel	88.89	16,000
Cost of cabin crew	6.11	1,100
Cost of flight	11.11	2,000
Depreciation	16.67	3,000
Aircraft maint	8.33	1,500
Total	<u>\$137.61</u>	<u>\$24,770</u>

Using this full cost analysis, it looks like Sunset will lose \$12.61 per passenger (or \$137.61 - 125.00). BUT, shouldn't only **RELEVANT** costs be considered? In other words, which costs can be **avoided** by choosing one option over another?

Relevant Costs

Almost all the costs are fixed with respect to the number of passengers on the plane. Therefore, the only relevant costs that would change between alternatives are the **variable** cost of food and drinks.

	Per Passenger	Per Round-trip
Cost of meals and drinks	<u>\$ 6.50</u>	\$ 1,170
Cost of fuel	88.89	16,000
Cost of cabin crew (four flight attendants)	6.11	1,100
Cost of flight crew	11.11	2,000
Depreciation of aircraft	16.67	3,000
Aircraft maintenance	8.33	1,500
Total	<u>\$137.61</u>	<u>\$24,770</u>

Excess Capacity Makes a Difference

In situations where excess capacity exists, the general rule is that the special order price must simply be higher than the additional variable costs incurred in accepting the special order, if excess capacity exists.

	EXCESS CAPACITY	NO EXCESS CAPACITY
Relevant costs:		
Meals	\$ 6.50	\$ 6.50
Opportunity costs from lost ticket revenue (Contribution margin lost)	0	\$ 268.50
Total relevant costs	<u>\$ 6.50</u>	<u>\$ 275.00</u>
Special-order ticket price	125.00	125.00
Profit (loss) from accepting special order	<u>\$118.50</u>	<u>\$(150.00)</u>

Excess Capacity Makes a Difference

If there is no excess capacity, Sunset will have to forgo the receipt of \$268.50 of contribution margin on each ticket (\$275 - \$6.50). By turning away full-pay customers, they will lose \$150. Therefore, they should not accept a special order for less than **\$275 per passenger.**

	EXCESS CAPACITY	NO EXCESS CAPACITY
Relevant costs:		
Meals	\$ 6.50	\$ 6.50
Opportunity costs from lost ticket revenue (Contribution margin lost)	0	\$ 268.50
Total relevant costs	<u>\$ 6.50</u>	<u>\$ 275.00</u>
Special-order ticket price	125.00	125.00
Profit (loss) from accepting special order	<u>\$118.50</u>	<u>\$(150.00)</u>

Fixed Costs Can Be Relevant

Fixed costs can be **relevant** to a special-order decision when they change, depending on the option chosen.

- Previously, four flight attendants were paid \$275 each and could serve the entire plane.
- What if one attendant was required for every 35 passengers?**
- If 170 passengers are flying as compared to the normal load of 140, that means, five attendants are needed, adding relevant costs of \$9.17 per passenger (or $\$275 \div 30$).

Relevant Costs Depend on the Situation

Assume Sunset has an additional airplane that is currently idle but can be chartered for the special order flight. **Will this change relevant costs?**

In this case, fuel costs, salaries of the flight and cabin crews, and maintenance are likely to be relevant, but depreciation is still not relevant.

Qualitative Factors to Consider

What if Passenger Load Predictions are Wrong?



What if Profitable Long-term Customers Turn to Competitors?

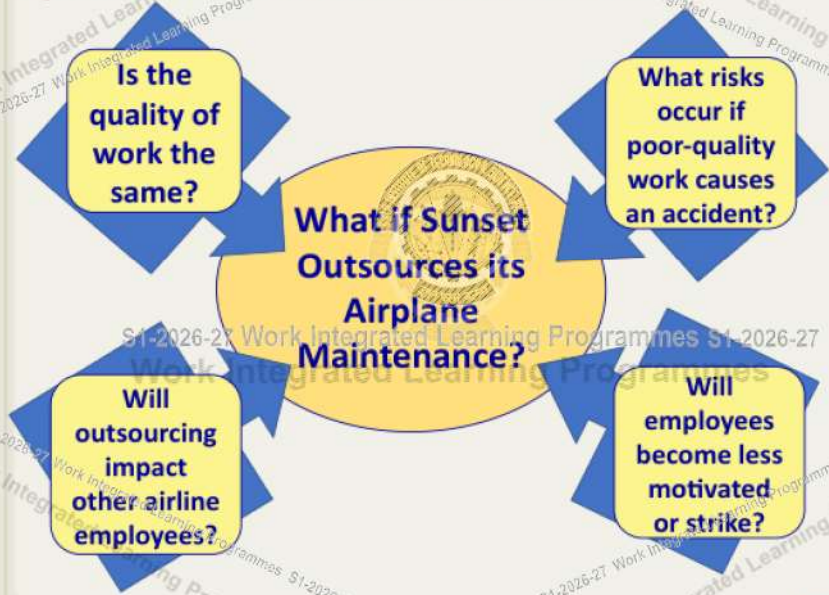
What is the Impact of Selling Discount Seats on Regular Fare Customers?

Outsourcing and Other Make-or-Buy Decisions

The Decision to Outsource

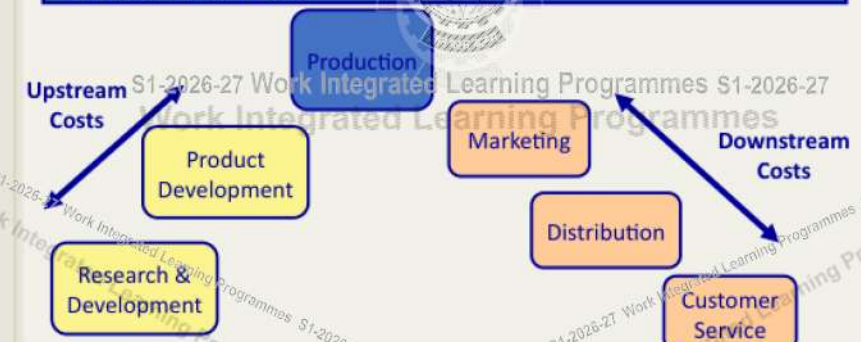
- Affects a wide range of organizations.
- Requires an in-depth analysis of relevant quantitative and qualitative factors, as well as costs and benefits and vertical integration

Qualitative Factors to Consider



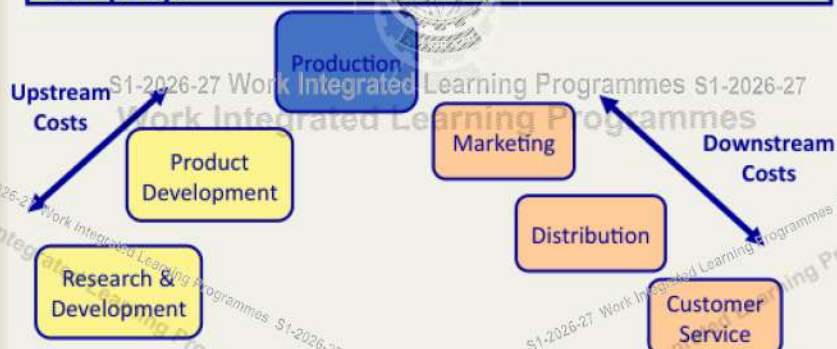
Vertical Integration

The value chain of an organization is simply the set of activities that increase the value of an organization's products and services. Vertical integration is accomplished when a company is involved in multiple steps of the value chain, extending from R&D to Customer Service.



Vertical Integration

Most companies operate with some form of vertical integration (they market the products they produce, or they develop the products they manufacture), but the extent of integration varies greatly from company to company and from product to product within a company.



Vertical Integration

Advantages: Not dependent on timely delivery and can control quality

Disadvantages: Suppliers may supply higher-quality items at less cost

Make-or-Buy Example

Birdie Maker wants to purchase 1,000 putters for its custom golf club sets from Ace Putters. Birdie currently makes the putters (costs shown below):



	Total (1,000 putters)	Per Unit
Direct materials	\$ 5,000	\$ 5.00
Direct labor	9,000	9.00
Variable manufacturing overhead	3,000	3.00
Fixed manufacturing overhead	9,500	9.50
Total cost	\$26,500	\$26.50

Make-or-Buy Decision

Ace Putters is offering to sell the putters to Birdie Maker for \$25 per putter. Since the fixed overhead of \$9.50 per unit is incurred regardless of the decision, the decision to outsource would cost the Birdie more than making the putters by \$8 per unit (or $8 \times 1,000 = \$8,000$ total).

	COST TO MAKE (PER UNIT)	COST TO BUY (PER UNIT)
Direct materials	\$ 5.00	
Direct labor	9.00	
Variable manufacturing overhead	3.00	
Fixed manufacturing overhead	9.50	
Purchase price from Ace Putters		\$ 25.00
	\$26.50	\$34.50

$\$34.50 - \$26.50 = \$8$

Make-or-Buy Decisions with Relevant Fixed Costs

Instead, assume that Birdie Maker will be able to return a specialized machine for putters to the lessor, saving \$5,500, if Ace makes the putters ($\$5,500 \div 1,000 = \5.50 per unit). This means that only **\$4.00** per unit will be **relevant** for the outsourcing decision.

	COST TO MAKE	COST TO BUY
Direct materials	\$ 5.00	
Direct labor	9.00	
Variable manufacturing overhead	3.00	
Fixed manufacturing overhead	9.50	
Purchase price from Ace Putters		\$ 25.00
	\$26.50	\$29.00

Make-or-Buy Decisions with Relevant Fixed Costs

Even though the cost difference shrinks to **\$2.50**, Birdie will still lose money. However, they also need to consider **qualitative factors** (like the quality of the putters, the importance of keeping up with changing technology, and the dependability of the supplier).

	COST TO MAKE	COST TO BUY
Direct materials	\$ 5.00	
Direct labor	9.00	
Variable manufacturing overhead	3.00	
Fixed manufacturing overhead	9.50	
Purchase price from Ace Putters		\$ 25.00
	\$26.50	\$29.00

The Make-or-Buy Decision with Relevant Opportunity Costs

Opportunity costs should also be considered in make-or-buy decisions. Assume that Birdie could rent out the factory space that is now used to manufacture putters for \$10,000. The outsourcing would save \$2.00 per unit.

	COST TO MAKE	COST TO BUY
Direct materials	\$ 5.00	
Direct labor	9.00	
Variable manufacturing overhead	3.00	
Fixed manufacturing overhead	9.50	\$ 9.50
Purchase price from Ace Putters		25.00
Rental of unused factory space		(10.00)
	\$26.50	\$24.50

The Decision to Drop a Product or Service

- The decision to drop a product or a service is among the most difficult that a manager can make.

Qualitative factors are sometimes more important than focusing solely on income.

Dropping a Product or Service



Clayton Herring is thinking about dropping the mud and snow (M&S) tires. They require more machine time to make than the other tires. Only \$2,000 of M&S's fixed costs are avoidable. Should they drop the line?

	Mud and Snow	All Other Tires	Total
Sales	\$25,500	\$150,000	\$175,500
Less: Direct materials	12,000	50,600	62,600
Direct labor	5,000	30,000	35,000
Variable overhead	2,000	12,000	14,000
Contribution margin	\$ 6,500	\$ 57,400	\$ 63,900
Less: Fixed overhead	7,000	21,000	28,000
Net income	\$ (500)	\$ 36,400	\$ 35,900

The Decision to Drop a Product

The \$5,000 of the fixed costs allocated to mud and snow tires would have to be reallocated to other product lines, because these costs remain even if the mud and snow tires are discontinued. If the company drops the line, it will lose money because the contribution margin decreases by \$6,500, whereas fixed costs decrease by only \$2,000 if the tires are dropped.

	WITH MUD AND SNOW TIRES	WITHOUT MUD AND SNOW TIRES	DIFFERENCE
Tire sales	\$175,500	\$150,000	
Less: Direct materials	62,600	50,600	
Direct labor	35,000	30,000	
Variable overhead	14,000	12,000	
Contribution margin	\$ 63,900	\$ 57,400	\$(6,500)
Less: Fixed overhead	28,000	26,000	2,000
Net income	\$ 35,900	\$ 31,400	\$(4,500)

Qualitative Factors

What impact will discontinuing the sale of mud and snow tires have on sales of the remaining product lines?

Clayton Herring may lose distributors, since they cannot provide a full line of tires.

Resource Utilization Decisions

A resource utilization decision requires an analysis of how best to use a resource that is available in limited supply.



Resource Utilization Decisions

Resource utilization decisions are typically short-term decisions.

Short-Run: Resources like machine time, labor hours, and shelf space are fixed and cannot be increased.

Long-Run: New machines can be purchased, more skilled laborers can be hired, and stores can be expanded.

Short-run constraints mean a focus on the **contribution margin** of each product per unit of limited resource rather than profitability of each product.

Scarce or Limited Resources



Assume Birdie Maker's limited resource is **machine time** when making golf balls. They produce 1 carton of the Pro Model in 30 minutes and 1 carton of the Tour Model in 45 minutes. If demand is not a factor, which model maximizes profit if machine time is limited to 300 hours?

	Pro Model	Tour Model
Sales price (per carton)	\$450	\$540
Less: Direct materials	200	265
Direct labor	50	50
Variable overhead	50	75
Contribution margin	<u>\$150</u>	<u>\$150</u>

The Resource Utilization Decision

Contribution margin per carton is the same for both the Pro Model and the Tour Model. However, the contribution margin per unit of limited resource results in each carton of Pro Model balls with a contribution margin of \$300 per hour of machine time, as compared to Tour Model balls at \$200 per hour of machine time.

	PRO	TOUR
Sales price (per carton)	\$450	\$540
Less: Direct materials	200	265
	50	50
	50	75
	\$150	\$150
÷ 0.50 hours	÷ 0.75 hours	
Contribution margin per machine hour	\$300	\$200

Birdie will maximize profit by selling only the Pro Model.

Optimal Product Mix

If demand for either model is limited, the company must decide on the optimal mix.

Assume demand is 400 cartons of Pro Model and 150 cartons of Tour Model.

Looking at the most profitable item per constraint, the capacity of Pro is 300 hours ÷ 0.5 hours = 600 cartons.

However, since Birdie can only sell 400 cartons, the optimal mix is 400 cartons of Pro (using 200 hours) and 133 cartons of Tour (100 hrs. ÷ 0.75 hrs.) using the remaining 100 hours.

Qualitative Issues and Other Options

Qualitative issues include:

- Is the visibility of the Tour ball on the professional tour a valuable source of advertising?
- Does it contribute to sales of the Pro Model?

Loosening the Constraints:

- Add machines to increase the amount of available
- Reduce the amount of machine time it takes to make a carton of balls.

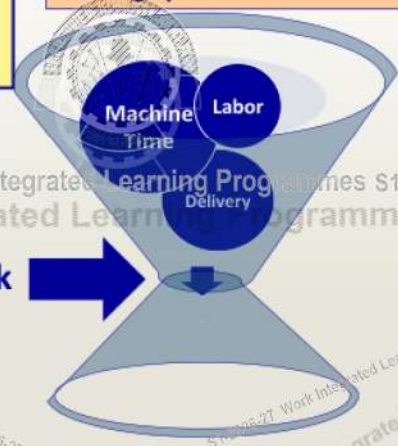
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The Theory of Constraints

- The theory of constraints is a management tool for dealing with constraints.

It identifies bottlenecks in the production process that limit throughput.

Bottleneck



Production Bottleneck for Birdie Maker



Sell or Process Further

Sell or Process Further

- Common management decision

Relevant Costs

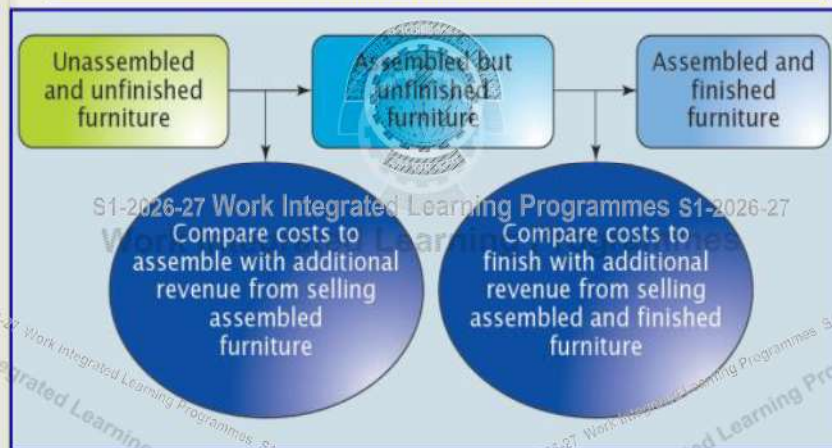
- Incremental or Additional Processing Costs

Decision Question

- Additional Revenue > Additional Costs, then Process Further
- If not, sell as is.

Sell or Process Further

Let's look at a furniture manufacturer that can sell furniture unassembled and unfinished, assembled and unfinished, or assembled and finished.



Sell or Process Further

Assume unassembled and unfinished tables cost \$100 to produce and can be sold for \$150. The company is considering selling assembled and finished tables for \$225 each. Additional assembly and finishing costs of \$45 per table would be required.

Incremental revenue = \$75
As long as the extra costs are less than \$75, the company will maximize profits by further processing.

Process further for a \$30 per table profit!

	Unassembled and unfinished tables	Assembled and finished tables	Incremental revenue and cost
Sales price	\$150	\$225	\$75
Cost to produce	100	145	45
Increase in income from further processing			\$30

Not relevant – sunk cost

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Thank You!

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Cost volume profit analysis

Buy or make
Capacity < outside India
Product Y + Product X

Cost-Volume-Profit (CVP) Analysis is a financial management tool that helps businesses understand the relationships between costs, sales volume, and profits. It is used to determine how changes in a company's cost structure (fixed and variable costs), sales price, and volume will affect its profitability.

Costs
Variable Cost / Marginal Cost
Fixed cost → Rent of Warehouse
50000
52000

Raw Material (RS)
5.00 501 1000
25000 25000 50000
Labour
50000 25000

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Cost volume profit analysis

Decisions of product profitability
Increase/Decrease production
Pricing decisions *impact of cost*
Accept new orders at lower than existing prices/not
Make a component in house or buy
Which product to produce when there is limited supply of raw material

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Cost volume profit analysis

$$\text{Sales} = \text{Profit} + \text{Fixed Cost} + \text{Variable Cost}$$

$$\begin{aligned} S &= P + FC + VC \\ S - V.C &= P + FC \\ \text{Profit} &= S - P = FC + VC \end{aligned}$$

$$\text{Profit} = S - P = FC + VC$$

Where S=Sales
V.C=Variable Cost

P=Profit
F.C=Fixed Cost

$$\text{Contribution} = SP - VC \text{ or } \text{Profit} + FC$$

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Cost volume profit analysis

•Total Contribution

•Total Contribution = Total Sales - Total Variable Costs

Per Unit Contribution (Cont pu)

Contribution per Unit = Selling Price per Unit - Variable Cost per Unit

$$\text{Contribution per Unit} = SP - V.C$$

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Sample ques 1

Eg.

A company produces 1,000 units of a product. The selling price per unit is Rs 80. The direct material cost per unit is Rs 20 and other variable costs per unit are Rs 30. The fixed costs are Rs 10,000. Find the

$$\text{Cont pu} = SP - VC = 80 - 50 = 30$$

$$\text{Total contribution} = 80 \times 1000 - 50 \times 1000 = 30000$$

PV ratio

$$\begin{aligned} \text{1000 units..} \\ SP &= 80 \\ VC &= 20 + 30 = 50 \\ FC &= 10,000 \end{aligned}$$

$$\text{Cont} = \frac{30000}{1000} \times 100 = 30\%$$

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Contribution Margin ratio/Profit volume ratio

$$\text{VC\%} + \text{PV RATIO \% (Contribution\%)} = 100\%$$



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BREAK EVEN POINT

A Ltd sells 300 units of product X in a year at Rs 100 per unit. The variable cost of the company is Rs 40 per unit and the fixed expenses are Rs 18000 per annum. Calculate the BEP of the company.

$$\begin{aligned} \text{BEP} &= \frac{\text{FC}}{\text{PV Ratio}} \\ \text{BEP (Per unit)} &= \frac{\text{FC}}{\text{Contribution per unit}} \\ &= \frac{\text{FC}}{\text{S.P.} - \text{V.C.}} \end{aligned}$$

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BREAK EVEN POINT

Break Even point is that level of sales at which the company earns no profit and no loss.

$$\text{BEP (Rs in Sales)} = \frac{\text{F.C.}}{\text{PV Ratio}}$$

$$\text{BEP (in units)} = \frac{\text{F.C.}}{\text{Contribution per unit}}$$

$$\text{BEP (RS, VALUE)} = \text{BEP (UNITS)} \times \text{SP PU}$$

$$\frac{\text{FC}}{\text{PV Ratio}} = \frac{\text{FC}}{\text{Contribution per unit}}$$

$$\times \text{SP/unit}$$

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BREAK EVEN POINT

$$\text{BEP} = \frac{\text{F.C.}}{\text{PV Ratio}}$$

$$\text{PV Ratio} = \frac{\text{S.P per unit} - \text{V.C per unit}}{\text{S.P per unit}}$$

$$= \frac{(100 - 40)}{100} \times 100$$

$$= \frac{60}{100} \times 100$$

$$= 60\%$$

$$\text{BEP} = 18000 / 60\%$$

$$= \text{Rs } 30000$$

$$\text{(or) } 300 \text{ units}$$

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BREAK EVEN POINT

Checking:

Sales(300*100) = 30000

Less: Variable Cost (40*300) = 12000

Contribution = 18000

Fixed Cost = 18000

Profit/Loss NIL

$$\text{Profit} + \text{FC} = \text{Sales} - \text{VC}$$

$$18000 + 18000 = 30000 - 12000$$

$$36000 = 18000$$

$$\text{Profit} + \text{FC} = \text{SP} - \text{VC}$$

$$18000 + 18000 = 30000 - 12000$$

$$36000 = 18000$$

$$\text{Profit} = 18000$$

$$\text{Profit} = 18000$$

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Margin of safety

is that portion of sales which contributes towards the profit of the company

Margin of Safety = Profit/PV Ratio

MS (in units) = Profit/Contribution Per Unit

Actual Sales = BEP Sales + Margin of Safety

$$\text{MS} = \text{Sales} - \text{BEP Sales}$$

$$\text{BEP} = \frac{\text{FC}}{\text{PV Ratio}}$$

$$\text{Profit} = \text{PV Ratio} \times \text{MS}$$

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MARGIN OF SAFETY

2) What is the Margin of safety and BEP of the company if the company earns a profit of Rs 24000? The variable cost is Rs 40 pu and the fixed cost is Rs 18000. Total units sold are 700 at Rs 100 pu.

Margin of safety = Profit/PV Ratio

= 24000/60%

= Rs 40000

$$\text{PV Ratio} = \frac{100 - 40}{100}$$

$$= \frac{60}{100}$$

$$= 60\%$$

Sales (in Rs) to earn a desired profit

Sales (in Rs) to earn a desired profit = (F.C. + Desired Profit)/PV Ratio

Sales (in units) to earn a desired profit = (F.C. + Desired Profit)/Contribution per unit

$$\text{Sales (in Rs)} = \frac{\text{F.C.} + \text{Desired Profit}}{\text{PV Ratio}}$$

$$\text{Sales (in Rs)} = \frac{\text{FC} + \text{DP}}{\text{PV Ratio}}$$

$$= \frac{18000 + 24000}{60\%}$$

$$= \frac{42000}{0.6}$$

$$= 70000$$

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Cost in Financial Statements



FORMULA 1 : CONT PU = SP PU - VC PU

FORMULA 2 : PV RATIO = CONT PU / SP PU X 100

FORMULA 3 : BEP (UNITS) = FC / CONT PU

FORMULA 4 : MS (UNITS) = PROFIT / CONT PU

MS (VALUE) = PROFIT / PV RATIO

FORMULA 5 : MS + BEP = TOTAL SALES

FORMULA 6 : VC% + PV RATIO % = 100%

Revenues	
Less: Cost of Goods Sold	
Gross Margin	
Less: Admn and S & D Expenses	
Operating Profit	

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Make or Buy



Make or buy

Export order

Constraining factors: linear programming

Toyota relies on outside suppliers to supply some parts and components but chooses to manufacture other parts internally.

The BBC outsources much of its accounting-based activities.

Saint Gobain supplies car glasses to all major car brands like Audi, BMW, Mercedes, Porsche, Skoda, Volkswagen, Volvo, Ford, Fiat, Honda, Hyundai, Toyota,

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Make or Buy

Often qualitative factors dictate management's make-or-buy decision.

Dell Computers must buy the Opteron chip for its personal computers from AMD because it does not have the know-how and technology to make the chip itself.

Sometimes, a company may prefer to make the product in-house to retain control of the product and technology. For example, in order to safeguard Coca-Cola's formula, the company does not outsource the manufacture of its concentrate.

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Special Orders

- 1) Do we have capacity to fulfil the special order?
- 2) Will the price asked by the buyer cover differential costs?
- 3) Are there any implications for fixed costs?
- 4) What qualitative issues might arise?

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Make or Buy

Do we have in house capacity?

Marginal costs of make vs buy?

Can the idle capacity be used to generate additional contribution if we buy?

Are there any implications for fixed costs?

What qualitative issues might arise?

All vc are relevant

FC are irrelevant unless otherwise mentioned

To calculate fc per unit ;fc should be divided by units manufactured not by capacity

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Thank You!

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CVP ANALYSIS

1) Determine the fixed expenses

Sales=240,000
Direct Materials=80,000
Direct Labour=50,000
Variable Overheads=20,000
Profit=50,000

2) From the following, calculate,

- Contribution per unit
- PV Ratio
- BEP in units and Rs
- What will be the selling price per unit if the break even point is brought down to 25,000 units?

Fixed Expenses 150,000
Variable cost per unit 10
Selling price per unit 15

3) From the following, calculate

- a) Break even point in amount of sales in Rs
b) Number of units that must be sold to earn a profit of Rs 120,000 per year
c) How many units must be sold to earn a net income of 15% on sales?

- Selling price per unit 40
- Variable manufacturing cost per unit 22
- Variable selling cost per unit 3

Fixed factory overheads 160,000
Fixed selling cost 20,000

4) Assume that the cost structure and selling price remain the same in Periods I and II. A company has the PV ratio is 20% sales of 140,000 and profit is 13,000 in one period. Calculate

- a) Fixed Cost
b) BEP for sales
c) profit when sales are Rs 100,000
d) Sales required to earn a profit of Rs 20,000
e) Margin of safety at a profit of Rs 15,000
f) Variable cost in period II

5) The PV ratio of a company dealing in precision instruments is 50% and the margin of safety is 40%. Work out the BEP and the net profit if the sale volume is Rs 50 lakhs.

6) The fixed costs amount to Rs 1,50,000 and the percentage of variable cost to sales is given to be 66.66%. If 100% capacity sales at normal capacity are Rs 900,000, find out the break even point and the percentage sales when it occurs. Find the margin of safety. Find the profit at 80% capacity sales.

7) The following information is given to you
Sales 400000
Fixed Costs 180000
Variable costs 270000

Calculate
PV Ratio
BEP
Ascertain by how much the value of sales must be increased for the company to break even

CVP ANALYSIS SOLUTIONS

1) $S - V = F + P$
Sales = 240000
 $V.C = 80000 + 50000 + 20000$
 $= 150000$
 $S - V = 240000 - 150000$

$\text{Contribution} = 90000$

$\text{Contribution} = 90000$
 $C = F \div P$
 $90000 = F \div 50000$
 $F.C = 40000$

2(i) $C = S - V$
 $= 15 - 10 = 5$

(ii) $PV \text{ Ratio} = (C/S) \times 100$
 $= (5/15) \times 100$
 $= 33.33\%$

(iii) $BEP = F.C / \text{Contribution}$
 $= 150000/5$
 $= 30000 \text{ units}$

(iv) $BEP \text{ (in Rs)} = BEP \text{ (in units)} \times SP \text{ per unit}$
 $= 30000 \times 15 = 450000$

(or) $F.C / PV \text{ Ratio} = 150000/33.33\% = 450000$

v) $BEP = F.C / \text{Contribution}$

Hence

$\text{Contribution} = F.C / BEP \text{ (in units)}$
 $= 150000/25000$
 $= 6 \text{ pu}$

$SP = \text{Contribution} + V.C \text{ pu}$
 $= 6 + 10 = 16 \text{ pu}$

3(i) $BEP = F.C / \text{Contribution pu}$
 $= (160000 + 20000) / (40 - 25)$
 $= 180000/15$
 $= 12000 \text{ units}$

ii) Output to earn a profit of Rs 120000 pa
 $= (F.C + \text{Desired Profit}) / \text{Contribution pu}$
 $= (180000 + 120000) / (40 - 25) = 20000 \text{ units}$

iii) Let Sales = X units

$X = (F.C + \text{Desired Profit}) / \text{Contribution pu}$
 $= (180000 + 15(40X)) / 15$
 $15X = 180000 + 6X$
 $9X = 180000$
 $X = 20000 \text{ Units}$

4) Assume that the cost structure and selling price remain the same in Periods I and II. A company has the PV ratio is 20% sales of 140,000 and profit is 13,000 in one period. Calculate

a) $F.C = (\text{Sales} \times PV \text{ Ratio}) - \text{Profits}$
 $= (140000 \times 20\%) - 13000$
 $= 15000$

b) $BEP = F.C / PV \text{ Ratio}$
 $= 15000/20\%$
 $= 75000$

c) $\text{Desired Profit} = (\text{Sales} \times PV \text{ Ratio}) - F.C$
 $= (100000 \times 20\%) - 15000$
 $= 5000$

d) Sales (in Rs) to earn a desired profit
 $= (F.C + \text{Desired Profit}) / PV \text{ Ratio}$
 $= (15000 + 20000) / 20\%$
 $= 175000$

$$\begin{aligned} \text{e) MS} &= \text{Profit/PV Ratio} \\ &= 15000/20\% \\ &= 75000 \end{aligned}$$

$$\begin{aligned} \text{f) V.C} &= 140000 - 15000 - 13000 \\ &= 112000 \end{aligned}$$

$$\begin{aligned} \text{5) Sales} &= 50,00,000 \\ \text{Less: Margin of Safety (40\%)} &= 20,00,000 \end{aligned}$$

$$\text{Sales at BEP} = 30,00,000$$

$$\text{PV Ratio} = 50\%$$

$$\text{M S} = \text{Profit/PV Ratio}$$

Hence

$$\begin{aligned} \text{M.S} \times \text{PV Ratio} &= \text{Profit} \\ 20,00,000 \times 50\% &= 10,00,000 \end{aligned}$$

$$\begin{aligned} \text{g) VC \%} &= 66.67\% \\ \text{So PV Ratio} &= 33.33\% \end{aligned}$$

$$\begin{aligned} \text{a) BEP} &= \text{FC/PV ratio} \\ &= 150,000/33.33\% \\ &= \text{Rs } 450,000 \end{aligned}$$

$$\begin{aligned} \% \text{ at which BEP sales occurs is } &= (450,000/900,000) \times 100 \\ &= 50\% \end{aligned}$$

$$\text{b) MS} = 900,000 - 450,000 = 450,000$$

$$\text{c) Profit at 80\% capacity}$$

$$\begin{aligned} 900,000 \times 80\% &= 720,000 \\ 720,000 \times 33.33\% &= 150,000 = 90,000 \end{aligned}$$

$$\text{Profit at 80\% capacity is Rs } 90,000$$

$$\begin{aligned} \text{7) pv Ratio} &= (400,000 - 270,000)/400,000 \times 100 \\ &= 32.5\% \end{aligned}$$

$$\text{BEP} = 180,000/32.5\% = \text{Rs } 553,846$$

$$\text{Sales must be increased by } 153,846 \text{ to break even } (553,846 - 400,000)$$

MAKE OR BUY

- 1) Fastide Cycle Ltd. purchases 20,000 bells p a from outside supplier at ₹ 10 each. The management feels that these be manufactured within the factory.

A machine costing ₹ 50,000 will be required to manufacture the item within the factory. The machine has an annual capacity of 30,000 units and life of 5 years. The following additional information are available:

Labour Cost per bell ₹ 3

Marginal level cost unit will be ₹ 4

Variable overheads = 100% of labour cost

You are required to advise whether:

- The company should continue to purchase the bells from outside supplier or should make them in the factory; and
- Should the company accept an order to supply 5,000 bells to the market at a selling price of ₹ 8.5 per unit?

- 2) A part No 293 used in the assembly of a product manufactured by your company has during the past 3 years been a bought out item. The current price of this part is Rs 120. Transportation costs amount to Rs 15 per piece. Sales tax is 10 % of the invoice price. Annual requirements of this part are 6000 units. Prepare a study to enable the management to reach a decision on whether to buy or manufacture this part. The following estimates are available per unit

	Rs
Raw Materials	96
Direct labour	8
Overheads at 800% of direct labour	64
Total Cost	168

In addition, special tools required to manufacture this product will have to be acquired at the cost of Rs 150,000. These are to be amortised over 5 years. The variable portion of the overhead is 100% of direct labour. Make your recommendations

- 3) A machine manufactures 10000 units of a product at a total cost of Rs 21 of which Rs 18 is variable. This part is available in the market at Rs 19. per unit. If the part is bought from the market, then the machine can be utilized to manufacture a component in the same quantity contributing Rs 2 per component or it can be hired out at Rs 21000. Recommend which alternative is profitable.

1)	
Material cost	4
Labour Cost	2
Variable overheads	2
TOTAL VC	8
ADD: FC PU	0.5
	8.5

Purchases cost of bell outside = ₹ 10

Annual savings in production of bells [20,000 bells (₹10 - ₹ 8.50)] = ₹ 30,000.

Calculation of depreciation per unit

Depreciation p.a. = ₹ 50,000/5 years = ₹ 10,000 p.a. Depreciation per unit = ₹ 10,000/20,000 = ₹ 0.50 p.u

- Since the marginal cost per unit of ₹ 8.5 is lower than the market price of ₹ 10, it is recommended to manufacture the component in the factory.
- Marginal cost per bell is Rs. 8.00. As depreciation of the machine is recovered on 20,000 bells, there will be no additional depreciation on the extra 5,000 bells to be sold in the market. Further the machine has additional capacity too. Therefore, the company is advised to supply 5,000 bells to the market at Rs. 8.50 per unit and make a profit of Re. 0.50 per unit i.e., total profit Rs. 2,500

2) ESTIMATED MANUFACTURING COST OF PART X-293.

	RATE PU	Total amount for 6000 units
Raw Material	96.00	5,76,000
Direct Labour	8.00	48,000
Variable overheads (100% of direct Wages)	8.00	48,000
Special tools, jigs and fixtures	5.00	30,000
	117.00	7,02,000

BUYING COST OF PART X-293

	RATE PU	Total amount for 6000 units
Current purchase price	120.00	7,20,000
Add: Sales tax	12.00	72,000
	132.00	7,92,000
Transportation costs	15.00	90,000
	147.00	8,82,000

It is recommended that part X-293 be manufactured as this gives a saving of Rs 180000 (882000-702000) p.a

3) ALTERNATIVE I

10000 units manufactured internally

Variable cost
= 10000 * 18 = Rs 180,000

ALTERNATIVE II

10000 units bought from outside, Capacity is used to manufacture another product.

Purchase Price (10000*19)	190,000
Less: Contribution from the other product (10000*2)	20,000
	170000

ALTERNATIVE III

10,000 units bought from outside, machine time is hired out.

Purchase Price (10000*19)	190000
Less: rent from hiring out	20000
	170000

Alternative III should be adopted as it is least expensive.

SPECIAL ORDER

1) The cost sheet of a product is given as under

	Rs	Rs PER UNIT
Direct materials		5.00

Direct wages		3.00
Factory overheads		
Fixed	25,000 (0.5 pu)	
Variable	0.50	1.00
Administrative expenses		0.75
Selling and distribution OH		
Fixed	12,500(0.25 pu)	
Variable	0.50	0.75 (0.5+0.25)
		10.50

The selling price is Rs 12 PU

The above figures are for 50,000 units. The capacity of the firm is 65000 units. A foreign customer wants to buy 15000 units at Rs 10 pu. Should the order be accepted? What would be your answer if the order was from a local merchant?

2) A company manufactured 10,000 units of a product at Rs 4 pu and there is a home market consuming the entire production at Rs 4.25 pu. In the year 2023 there is a fall in the demand in the home market which can consume 10,000 units at a price of Rs 3.72 pu. The total cost for 10,000 units is as follows:

Direct Material	15,000
Direct Wages	11,000
Fixed OH	14,000
Variable OH	6,000

It is realized that the foreign market can consume 20,000 units if offered at a price of Rs 3.55 pu. These 20,000 units will be over and above the 10,000 units being sold in the domestic market. It is also discovered that for every additional 10,000 units over the initial 10,000 the fixed OH will increase by 10%. Is it worthwhile to capture the foreign market?

SPECIAL ORDER SOLUTIONS

1) MARGINAL COST OR ADDITIONAL COST FOR 15000 UNITS

	Rs PU	Rs (per 15000 units)
Direct Materials	5.00	75,000
Direct Labour	3.00	45,000

Prime Cost	8.00	1,20,000
Variable overheads		
Factory	0.50	7,500
Selling and distribution	0.50	7,500
Marginal Cost	9.00	1,35,000
Sales	10.00	1,50,000
	1.00	15,000

The order from the foreign customer will give an additional contribution of Rs 15000. Hence it should be accepted.
The order from the local merchants should not be accepted at a price of Rs 10 as it is less than the normal price (Rs 12). This might affect relationships with other customers and there might be a general tendency of reduction in price.

2) Analysis of cost data for 10000 units

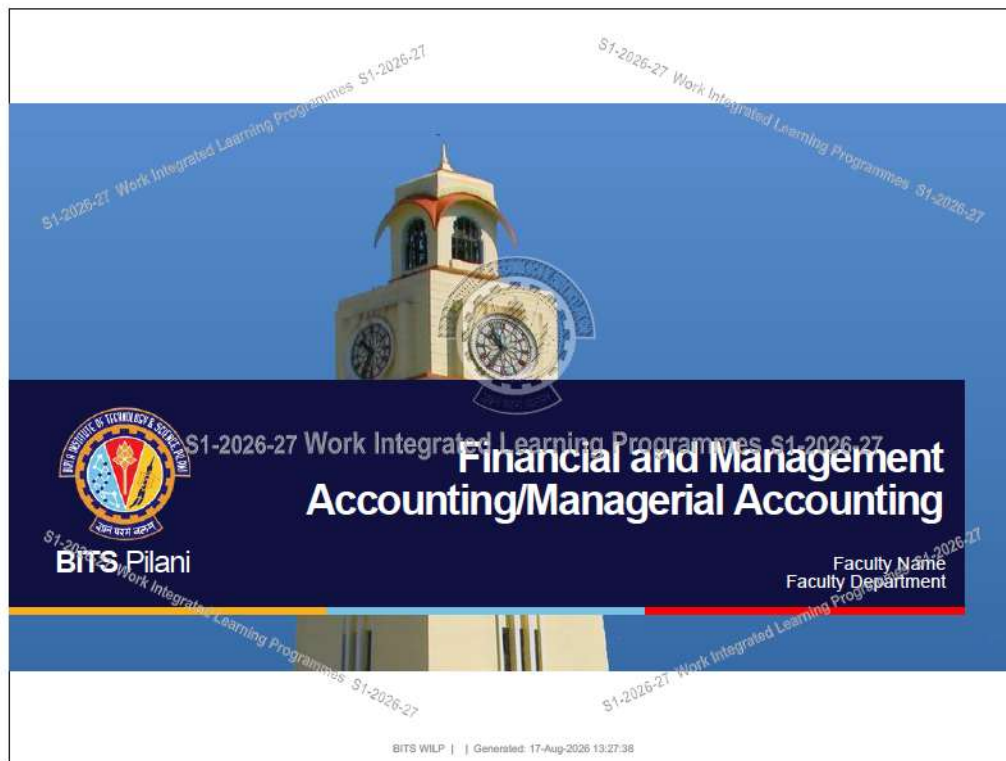
	Rs PU	Total
Direct Material	1.50	15000
Direct Labour	1.10	11000
Variable overheads	0.40	4000
Marginal Cost	3.20	32000
Fixed Overheads	0.80	8000
Total Cost	4.00	40000

Statement showing the advisability of selling in the foreign market

	Sale – 10000 units	Sale – 20000 Units	TOTAL
SALES (10000*3.72)			
(20000*3.55)	37,200	71,000	1,08,200

Direct Materials	15,000	30,000	45,000
Direct Labour	11,000	22,000	33,000
Variable overheads	6,000	12,000	18,000
Marginal Cost	32,000	64,000	96,000
CONTRIBUTION	5,200	7,000	12,200
Less: Fixed Overheads	8,000	1,600 (800+800)	9,600
PROFIT/LOSS	(2,800)	5,320	2,520

Hence 10000 units should be sold in the home market at 3.72 pu and 20000 units should be sold in the foreign market at Rs 3.55 pu. The total profit from this is Rs 2600.



Standard cost is the ^{estimated} predetermined cost based on a technical estimate of materials, labour and overhead for a selected period of time and for a prescribed set of working conditions. Standard costs help us to calculate variances which help in controlling costs and finding reasons for deviations.

direct cost
→ Max is one year or less than 1 year

Planned cost - Actual cost = Variances.

Control → correct measure

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Material Cost variance: (MCV)

Std cost of material - actual cost of material

Std cost of material = $SQ \times SP$

Actual cost of material = $AQ \times AP$

Standard cost

AQ=Actual Quantity AP=Actual price per unit

SQ=Standard quantity SP= Standard price per unit

Note: SQ is the standard qty of material for actual output ie if the standard quantity of material is 5 kg for 1 unit of output and actual output is 200 units, then $SQ = 5 \times 200 = 1,000$ kg

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Material Price Variance (MPV)
 $AQ(SP - AP)$

Material Usage Variance (MUV)
 $SP(SQ - AQ)$

Note: SQ is the standard qty of material for actual output

Check : $MPV + MUV = MCV$

Material cost Variance.

Material Price Variance

Material Usage Variance

$SP(SQ - AQ)$

$AQ(SP - AP)$

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Labour Variances

Labour cost variance (LCV)

Std cost of labour – Actual cost of labour

Std cost of labour = $ST \times SR$

Actual cost of labour = $AT \times AR$

$$LCV = \underline{ST \times SR} - AT \times AR$$

Labour rate variance: $AT(SR - AR)$

Labour efficiency variance: $(ST - AT) \times SR$

*ST is the standard time for actual production

$$LRV + LEV = LCV$$

Handwritten diagram showing the breakdown of LCV:

$$LCV = \underbrace{AT(SR - AR)}_{\text{Rate LRV}} + \underbrace{SR(ST - AT)}_{\text{Efficient LEV}}$$